Math Rock Essentials

数学摇滚精髓

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翻译说明 / Translator's Note

> **ENG** This is a Chinese translation of **Math Rock Essentials** by Stephen Hazel. The original text is presented alongside the translation for reference. Any translation errors are my own.

> **译** 本文是 Stephen Hazel 所著 **Math Rock Essentials** （数学摇滚精髓）的中文翻译版。原文与译文并列呈现，以供对照参考。翻译如有错漏，概由译者负责。

排版惯例

深蓝色标示原文，暗红色标示译文。紫色文字为术语并列有英文原文。

Part I

Math Rock Guitar Techniques

数学摇滚吉他技巧

How to Practice? / 如何练习

> ENG Before we dive into the core math rock guitar techniques, I would like to share an effective practice strategy. This approach is aimed at learning new techniques or refining existing ones. It may sound counterintuitive, but it involves intentionally making mistakes. This process of generating errors is essential for skill improvement. Here is how it works:

> 译 在深入学习核心数学摇滚吉他技巧之前，我想分享一个高效的练习策略。这个方法旨在学习新技术或打磨已有技巧。听起来可能反直觉，但它需要你有意地制造错误——产生错误的过程是提升技能的关键。具体方法如下：

> ENG **1. Generate Errors:** The brain processes errors after each practice session. When you return to practice, you will begin to see improvements in that skill. Therefore, aim to perform as many repetitions as possible of the technique you are practicing within your dedicated practice time.

> 译 **1. 制造错误：**大脑会在每次练习后处理错误信息。当你再次练习时，你会发现在该项技能上有所进步。因此，在练习时间内，尽量多做你正在练习的技巧的重复。

> ENG **2. Use a Metronome:** A metronome should be your ally in this process. Start with a tempo where you can comfortably play the exercise, then gradually increase it (typically by 3 bpm increments) until you reach a point where you are no longer comfortable. This is the sweet spot for improvement because you will be generating more errors in your repetitions, pushing your brain's neural plasticity to adapt to this new challenge.

> 译 **2. 使用节拍器：**节拍器应该成为你的好帮手。从一个你能舒适演奏练习的速度开始，然后逐渐加快（通常每次增加 3 BPM），直到你感到不再舒适为止。这就是进步的最佳点——因为你在重复中产生了更多错误，推动大脑的神经可塑性去适应新的挑战。

> ENG **3. Focused Practice:** When practicing a technique (skill), it is crucial to have a specific focus, even if it is just one aspect of the technique. For instance, you might dedicate a practice session to observing the movement of one of your hands or focusing on the positioning of one of your hands. By rotating your focus in different practice sessions, you can systematically improve various aspects of your technique.

> 译 **3. 专注练习：**练习一项技巧时，有一个明确的专注点至关重要，哪怕只是该技巧的一个方面。例如，你可以用一次练习专门观察某只手的运动，或专注于某只手的定位。通过在每次练习中轮换关注点，你可以系统性地提升技巧的各个方面。

> ENG In summary, I encourage you to try this practice strategy for the upcoming techniques and exercises. After each practice session, take a few minutes to allow your brain to simply reflect on what you have practiced. Good luck and may your practice sessions be productive and rewarding!

> 译 总之，我鼓励你在接下来的技巧和练习中尝试这个策略。每次练习后，花几分钟让大脑沉淀一下刚刚练习的内容。祝你好运，愿你的每次练习都高效而充实！

1 Finger Tapping / 点弦

> **ENG** Finger tapping is considered a staple of math rock guitar techniques by many. It is widely employed in the style, and in similar styles that are often attributed to math rock (Midwest emo, math core, neo soul, Nu Prog and so on). The technique is employed somewhat differently when compared to more conventional forms of finger tapping, and this difference is what makes it unique to mathy styles of guitar playing.

> **译** 点弦被许多人认为是数学摇滚吉他技巧的标志。它在这一风格以及与数学摇滚相关的类似风格中被广泛使用（例如 Midwest emo、math core、neo soul、Nu Prog 等）。与更传统的点弦方式相比，数学摇滚点弦的使用方式有所不同，而这种差异恰恰使它在“数学”吉他演奏中独树一帜。

> **ENG** I can only speculate on its rise in prominence, perhaps it is due to wanting to play rhythm and lead guitar at the same time, or it could simply just be to look flashy. Either way, it can sound excellent when done right and look hella (pun intended) impressive too! It has developed into being as synonymous with the style as much as the use of odd time signatures has come to be.

> **译** 我只能猜测它走红的原因。也许是吉他手希望同时演奏节奏和主音，或许仅仅是为了看起来更炫。不管怎样，只要演奏得当，它听起来就能非常出色，而且看起来也很惊艳！它已经如同奇数拍号一样，成为这种风格的代名词之一。

> **ENG** Here is a list of a few guitarists whom of which regularly employ the technique in their writing and playing:

> **译** 下面列举了一些在创作和演奏中经常使用点弦技术的吉他手：

- > **ENG** Yvette Young (Covet)

  > **译** Yvette Young（Covet）

- > **ENG** Tim Collis (TTNG)

  > **译** Tim Collis（TTNG）

- > **ENG** Brock Benzel (Invalids)

  > **译** Brock Benzel（Invalids）

- > **ENG** Dylan Mattheisen (Tiny Moving Parts)

  > **译** Dylan Mattheisen（Tiny Moving Parts）

- > **ENG** Marcos Mena (Standards)

  > **译** Marcos Mena（Standards）

- > **ENG** Tim Henson (Polyphia)

  > **译** Tim Henson（Polyphia）

- > ENG Nick Reinhart (Tera Melos)

  > 译 Nick Reinhart（Tera Melos）

> ENG In the following section, you will find exercises, technique pointers, and handy tips to get you on your way to tapping like a mathematical pro!

> 译 在接下来的部分，你会找到练习、技术提示和实用建议，帮助你像数学摇滚高手一样掌握点弦！

Finger Tapping Technique Types Explored > ENG For this segment we will be exploring two handed tapping and tapping with chords.

> 译 本节我们将探索双手点弦与和弦点弦两种技巧。

> ENG Tablature key so you know which hand to be using for tapping in each exercise:

> 译 下面的吉他谱符号将告诉你在每个练习中哪只手负责点弦：

- > ENG A symbol indicates tapping with your fretting hand (hammer-on from nowhere).

  > 译 某种符号表示由按弦手点弦（从空中敲击产生声音）。

- > ENG A different symbol indicates tapping with your strumming hand.

  > 译 另一种符号表示由弹弦手点弦。

Finger Tapping Technique Pointers > ENG Evenness of tapping: try to keep volumes even as possible at first. Later, you can experiment with dynamics in your tapping, but first I recommend trying to be as dynamically even as possible because this will improve your control.

> 译 点弦音量均衡：刚开始时尽量保持音量一致。之后你可以尝试在点弦中加入力度变化，但我建议先以动态均匀为目标，因为这会提升你的控制力。

Developing finger strength balance > ENG To achieve better finger strength balance, I recommend dedicating more time to your ring and little fingers on both hands. The exercise below is an excellent starting point for improving finger evenness and building strength.

> 译 为了获得更好的手指力量平衡，我建议双手都多练习无名指和小指。下面的练习是提高手指均衡性和力量的绝佳起点。

1. > ENG Position your fingers on the fifth to eighth frets of the fretboard.

   > 译 将手指放在指板第 5 到第 8 品。

2. > ENG Use your fretting hand to tap the strings in this order: index, middle, ring, and little finger.

   > 译 用按弦手按顺序点弦：食指、中指、无名指、小指。

3. > ENG Move to the ninth to twelfth frets and use your strumming hand to tap in the same finger order.

   > 译 移到第 9 到第 12 品，用弹弦手按同样的指序点弦。

4. > ENG A useful tip for your little finger on your strumming hand is to place your ring finger on top of it (apply pressure to your little finger's nail) to add more strength to the tap.

   > 译 对于弹弦手的小指，一个实用技巧是让无名指覆盖在它上面（对小指指甲施加压力），以增强点弦力度。

5. > ENG Once you feel comfortable on one string, try moving up and down all the strings.

   > 译 当你在一根弦上已经熟练后，尝试在所有弦上上下移动练习。

6. > ENG Ensure you tap each note for the specified duration (eighth notes: count 1 & 2 & 3 & 4 &), and make sure each note sounds correctly.

   > 译 确保你每个音都按照指定时值点出（八分音符：数 1 & 2 & 3 & 4 &），并保证每个音都能清晰发声。

7. > ENG Do not forget to use a metronome to help with timing and consistency.

   > 译 别忘了使用节拍器，它能帮助你保持节奏和稳定。

Finger Strength & Dexterity Development > ENG As you practice finger tapping, your tapping dexterity will naturally improve. To get you started though, I suggest the following exercises. These also double up as excellent warm-up routines!

> 译 随着你练习点弦，你的灵活性会自然提高。然而为了帮助你起步，我建议先做以下练习，它们也非常适合作为热身。

> ENG The following exercise is adapted from a John Petrucci exercise a guitar teacher showed me years ago. It has some good stretches and notice how the exercise becomes inverse of itself as it progresses and arrives back at the original fingering at the beginning of the repeat. I have added which fingers you should be using next to each note head in the exercise. Here is a key for which numbers refer to which finger on your fretting hand:

> 译 下面的练习改编自多年前一位吉他老师给我的 John Petrucci 练习。它包含了很好的伸展动作，并且随着进行会逐渐倒转，最终回到重复开始时的指型。我在谱例旁边标注了每个音符应使用的手指。以下是按弦手手指对应的数字说明：

> ENG Tip: when tapping, use the underside of your index finger to mute unwanted string ring (when you are not using it to tap of course!) by lightly touching strings that could potentially ring. To clarify, you are not to push down on the strings, just apply a light pressure for it to be enough to stop unwanted strings from ringing out. This is something you should develop if you have not already because it will make your tapping sound much cleaner. Also, when you are tapping on higher strings, G, B, or E for example, you should lightly rest the palm of your strumming hand on the lower strings (E, A, D) to mute them.

> 译 提示：点弦时，用食指的下侧轻触不需要发声的弦来阻止它们共鸣（当然不是按下，只需轻触即可）。这会让你的点弦听起来更干净。如果你在高音弦上点弦，例如 G、B、E，可以让弹弦手的手掌轻轻搁在低音弦（E、A、D）上以制音。

> ENG For our strumming hand, we have the same exercises but higher up the fretboard as this is more common territory for that hand. It will feel awkward at first, and you will probably hit some bum notes, nevertheless, persevere, it will get better, trust me!

> 译 对于弹弦手，我们也有一套类似的练习，只不过位置更靠近指板的高把位，因为这是弹弦手更常见的活动区域。一开始会感觉不自然，也可能会击到一些哑音，但坚持下去，它会变得更好，相信我！

Hand/Wrist Positioning for Finger Tapping > ENG To mitigate any unnecessary strain on your fingers, hands, and wrists I have a few tips to help!

> 译 为了减轻手指、手和手腕的不必要压力，我有几个建议可以帮助你！

> ENG First, try to keep your fretting hand's wrist straight as possible, it is not always possible but aim to keep it so:

> 译 首先，尽量保持按弦手的手腕尽可能笔直。这并不总是可行，但应作为目标。

> ENG Lastly, for your strumming hand, try to get your hand in-line with your fretboard (see left picture below for clarification) so that your hand and fingers are facing towards the nut of your guitar. You will find this more comfortable and easier to play than having you hand and fingers perpendicular to the fretboard. I like to anchor my thumb on the top side of the fretboard when finger tapping for stability, but this is not always an option (see the picture on the right below for clarification).

> 译 最后，对于弹弦手，尽量让手与指板保持同一直线（手和手指朝向琴枕）。你会发现这比让手和指板垂直更舒适，也更容易弹奏。我喜欢在点弦时将大拇指固定在指板上方以增加稳定性，但这并非总是可行。

Two Handed Tapping & Alternating Finger Tapping > ENG Two handed tapping mainly consist of alternating tapping (alternating between each hand when tapping). This is the most prominent version of tapping employed by math rock guitarists.

> 译 双手点弦主要由交替点弦构成（两只手交替点弦）。这是数学摇滚吉他手最常使用的点弦形式。

> ENG Occasionally, you might encounter a situation where you need to tap notes simultaneously with both hands, which, in my opinion, can be quite challenging! For instance, when attempting this section in my finger-tapping cover of "Santa Claus is Coming to Town," I personally find it more difficult to coordinate tapping with both hands simultaneously. A helpful tip to make this exercise easier is to first learn each part separately and then gradually combine the two.

> 译 偶尔你可能会遇到需要两只手同时点弦的情况，这在我看来相当具有挑战性。例如，在我翻弹《Santa Claus is Coming to Town》的点弦部分时，我个人觉得同时协调两只手更难。一个实用技巧是先分别学习每个声部，然后再逐渐把它们合在一起。

> ENG Below, you will find a bunch of exercises aimed at improving your two-handed tapping technique. In later sections, we will explore chords, arpeggios, and scales to create finger tapping ideas, but for now let's concentrate on getting better at the technique.

> 译 下面你会找到一系列旨在提升双手点弦技巧的练习。后续章节我们会探索和弦、琶音和音阶以创造点弦素材，但现在我们先集中练好基础技巧。

> ENG Exercise 1.E goes easy on the strumming hand and has a small ascending and descending pattern in it.

> 译 练习 1.E 对弹弦手比较友好，包含一个小的上行和下行模式。

> ENG Exercise 1.F is built around a major seven arpeggio. It starts with two notes and adds an extra note in each bar that follows. This is a good exercise to demonstrate simply how you can break an arpeggio into parts for tapping ideas.

> 译 练习 1.F 以大七琶音为基础。从两个音开始，每个小节加入一个额外音。这个练习很好地演示了如何将琶音拆分成点弦素材。

> ENG Exercise 1.G is based around extended seven chord arpeggios. I have split each arpeggio between both hands. As each bar progresses a new arpeggio of the major scale is introduced. Take your time with each bar, then glue it all together when you are comfortable with each individual arpeggio.

> 译 练习 1.G 基于扩展七和弦琶音。我将每个琶音分配给双手。随着每个小节的进行，会引入一个新的大调琶音。先慢慢练习每个小节，熟练后再把它们连起来。

> ENG I have included the following two exercises (NS1 & NS2) as next step exercises. They are more like the type of tapping you are to come across, but they still have an exercise feel to them. Never-the-less, I recommend them as the next step to take, and they will set you up nicely to attempt the test exercises at the end of this section.

> 译 我将以下两个练习（NS1 和 NS2）作为下一步练习。它们更接近你实际会遇到的点弦形式，但仍保留练习的感觉。无论如何，我建议你把它们作为下一阶段，并以此为基础去挑战本节末尾的测试练习。

> ENG Test Yourself: Two Handed Finger Tapping Exercises in Context

> 译 自测：双手点弦在音乐语境中的练习

> ENG As you have reached the end of the two-hand finger tapping exercises, take a moment to reflect on your progress so far. Challenge yourself by learning the following two exercises. By mastering these, you are already ahead of many other guitarists who either have not learned or have put minimal effort into finger tapping. Pause for a moment, and give yourself a well- deserved pat on the back, you have earned it.

> 译 当你完成双手点弦练习后，花一点时间回顾一下迄今为止的进步。挑战自己，学习下面这两个练习。只要掌握它们，你就已经领先于许多尚未学习或付出很少努力的吉他手了。停下来，给自己一个应得的肯定，你配得上这份成就。

> ENG To round off the two-hand tapping segment, I have listed a bunch of songs that use two- handed tapping at some point within the song. They are a great way to expand your tapping repertoire, and through osmosis, you will start to include things that you learn into your own playing and song writing through already established math rock content.

> 译 为了结束双手点弦部分，我列出了一些在歌曲中某个时刻使用双手点弦的作品。它们是扩展点弦曲目库的好途径，通过潜移默化，你将把学到的东西融入自己的演奏和创作之中。

- > ENG Maps & Atlases —Everyplace is a house (Beginner)

  > 译 Maps & Atlases —Everyplace is a house（入门）

- > ENG Tiny Moving Parts —Always Focused Intro (Intermediate)

  > 译 Tiny Moving Parts —Always Focused Intro（中级）

- > ENG Don Cabellero —Fire Back About Your New Baby's Sex (Intermediate)

  > 译 Don Cabellero —Fire Back About Your New Baby's Sex（中级）

- > ENG Tera Melos —Last Smile for Jaron Intro (Intermediate)

  > 译 Tera Melos —Last Smile for Jaron Intro（中级）

- > ENG Tera Melos —40 Rods to The Hogs Head Intro (Advanced)

  > 译 Tera Melos —40 Rods to The Hogs Head Intro（高级）

- > ENG Covet —Pelagic Intro (Advanced)

  > 译 Covet —Pelagic Intro（高级）

- > ENG Standards —Kale & Strawberries Intro (Advanced)

  > 译 Standards —Kale & Strawberries Intro（高级）

Finger Tapping with Chords > ENG This technique is basically what the name implies; strum or arpeggiate a chord and tap notes along with it. You could also fret a chord or a particular set of notes, but not pick or strum the shape, and instead tap notes and pull off letting the fretted note or notes of your chord ring out underneath.

> 译 这种技巧基本上就是其名称所指：拨响或琶音和弦，同时加入点弦。你也可以按住一个和弦或特定音型，不拨弦，而是点弦并拉开，让按住的音符在下方继续共鸣。

> ENG This is my favourite tapping technique. It is particularly useful for bands with one guitarist because it fills out the sound by potentially playing harmony and a melody at the same time. The technique mainly consists

of tapping single notes, or combinations of two, and sometimes, though rarely, three notes at the same time then pulling off, which lets the notes of the chord you are fretting ring out as if they were open notes. There are no hard rules on which fingers to use for tapping, but generally you will feel most comfortable using your index and middle fingers when not using a pick, and if you are using a pick, try your middle and ring fingers.

> 译 这是我最喜欢的点弦技巧。对于只有一个吉他手的乐队特别有用，因为它可以同时演奏和声与旋律，丰富声音。该技巧主要包括点单音、两个音的组合，有时（虽少见）还会同时点三个音，然后拉开，让按住的和弦音像开放音一样持续响起。点弦并没有硬性规定使用哪些手指，但通常不使用拨片时更习惯用食指和中指，使用拨片时可以尝试用中指和无名指。

> ENG Here is an exercise to get you started. Hold an A major barre chord, strum the chord and then use two fingers to tap and pull off letting the notes of the A major chord ring out underneath.

> 译 下面是一个入门练习。按住 A 大调把位，拨响和弦，然后用两个手指点弦并拉开，让 A 大调和弦的音在下方持续共鸣。

> ENG For exercise 1.L, I adapted an exercise from the amazingly talented guitarist Ichika Nito. For this exercise, I recommend practicing the chord changes first (I have listed them above the exercise for clarity), and then attempt the tapping parts. This way you are not trying to master two things at once, which will make the exercise more manageable. Remember to anchor your thumb on the side of the fret board. Then, when you are comfortable, try doing the same exercise with a pick and tap the tapped parts with your middle and ring finger.

> 译 对于练习 1.L，我改编自天才吉他手 Ichika Nito 的练习。建议先练习和弦变化（我在练习上方列出了清晰的和弦），然后再尝试点弦部分。这样你不会同时掌握两件事，练起来更容易。记得将大拇指固定在指板侧面。熟练后，可以尝试使用拨片，并用中指和无名指点弦。

> ENG By practicing both tapping approaches, you will develop your tapping technique for both hybrid picking and fingerpicking techniques. Lastly, notice that it is in 5/4 time, so there are five beats instead of the usual four, and notice you are tapping out a triplet rhythm on each beat after strumming each chord.

> 译 通过练习这两种点弦方式，你将同时提升混合拨弦和指弹点弦技巧。最后请注意，练习采用 5/4 拍，因此每小节有五拍而不是通常的四拍，并且每次拨完和弦后，你会用三连音节奏来点弦。

> ENG For exercise 1.M, fret and hold the chord shapes then tap and pull-off. The tapped notes are octaves of each chord. This octave higher tapping method is one option you can employ for finger tapping with chords, but it in context it will be better to mix it up with other notes to make it sound less like an exercise. Again, I have included the chord shapes for clarity:

> 译 对于练习 1.M，按住和弦形状然后点弦并拉开。点弦音是每个和弦的同名八度。这种高八度点弦方法是和弦点弦的一种方式，但在乐句中最好与其他音混合使用，避免听起来太像练习。再次说明，我已为你提供了和弦形状以便理解。

> ENG Test Yourself: Finger Tapping with Chords Exercises in Context

> 译 自测：和弦点弦在语境中的练习

> ENG Lastly, let us conclude this section with two examples of how finger tapping with chords can be used in context.

> 译 最后，让我们用两个示例来结束本节，展示和弦点弦如何在语境中使用。

2 Fingerpicking / 指弹

> ENG Fingerpicking is an incredibly valuable technique that helps simplify intricate picking parts and makes it easier to play complex pieces involving frequent string skipping. Furthermore, this approach lends a distinct character to our guitar playing, making it suitable for clean tones or acoustic guitars, introducing unique timbres to our playing. Notably, in the realm of math rock, guitarists like Tim Collis of TTNG and Yvette Young of Covet are masters of this technique.

> 译 指弹是一项非常宝贵的技巧，它能简化复杂的拨奏部分，并使得频繁跳弦的复杂乐句更容易演奏。此外，这种演奏方式还能为吉他带来独特的质感，尤其适合干净音色或木吉他，能够丰富演奏音色。在数学摇滚领域，TTNG 的 Tim Collis 和 Covet 的 Yvette Young 就是这项技巧的高手。

> ENG There is a wealth of creative possibilities awaiting exploration with this method, so let's dive right in.

> 译 这种方法蕴含了大量的创作可能性，下面我们就开始深入探索。

Developing Dexterity & Finger Strength > ENG To begin with, I recommend resting the palm of your picking hand lightly on the bridge of your guitar so that your fingers are poised above the strings and ready to pluck them. Your hand should not have to move much at all because your fingers should be spaced out enough to cover all the strings adequately. If you feel strain, perhaps you are rushing, not warming up enough, or positioning your wrist incorrectly. Your hand and wrist should feel relaxed, so if they are not, you may be doing something incorrectly, so try to address this as you work your way through the following exercises.

> 译 首先，我建议将拨弦手的手掌轻轻放在琴桥上，让手指悬在琴弦上方，随时准备拨弦。手不应移动过多，手指应足够分开以覆盖所有琴弦。如果你感到紧张，可能是你太急了、热身不够，或者手腕位置不对。手和手腕应保持放松，否则你可能做错了什么，练习时请注意修正。

> ENG To begin with, exercise 2.A is an addition exercise where an extra finger is introduced as the exercise progresses. Once you are comfortable with playing it, try it backwards too.

> 译 首先，练习 2.A 是一个加法练习，随着练习进行会逐渐加入额外的手指。当你熟练后，也尝试倒着练习。

Fingerpicking Patterns > ENG When you observe math rock guitarists who frequently use fingerpicking, you will begin to notice a common pattern: they often begin their riffs and licks from a lower note on the guitar, typically this will be the root note of a chord. Sometimes, they let this note ring out, especially if it is an open string. Other times, they do not. This approach is used to create a sense of harmony within the overall musical idea, suggesting harmonic movement in the progression, if there is one. Typically, this is followed by an arpeggio that captures the essence or "feel" of the underlying chord. For example, if we look at the following TTNG inspired idea. The circled lower notes in red show where each musical phrase is built from, and the notes that follow each circled note make up an arpeggiated chord and in turn, an arpeggiated melody of sorts.

> 译 当你观察经常使用指弹的数学摇滚吉他手时，会注意到一个共同点：他们常常从吉他上的低音开始乐句，通常是和弦的根音。有时他们会让这个音持续响起，尤其是当它是空弦时；有时则不会。这种做法可以在整体音乐理念中营造和声感，暗示和弦进行的走向。通常接下来会出现一个琶音，体现出和弦的本质或“感觉”。例如下面这个受 TTNG 启发的例子。红色圈出的低音显示了每个乐句的起点，随后出现的音符构成了琶音和旋律。

> ENG Exercises 2.B, 2.C, and 2.D are beneficial for getting started with picking patterns you are likely to come across, albeit in a somewhat simplified manner. Exercise 2.B adds a finger for each bar, exercise 2.C is designed to practice some basic picking patterns with three fingers and a thumb (which I find is more commonly used than all fingers and a thumb), and lastly exercise 2.D makes use of the same patterns but includes the little finger. Notice the change to 5/4 time to fit in the extra finger. Remember, precision over speed is vital, and use a metronome! Once you are comfortable with playing these picking patterns, try applying them to some chord shapes you like to use.

> 译 练习 2.B、2.C 和 2.D 是很好的入门模式，虽然它们相对简化，但非常接近你会遇到的实际指弹模式。练习 2.B 每小节增加一个手指，练习 2.C 的设计是用三指加拇指练习基础拨弦模式（我发现这种方式比所有手指加拇指更常见），而练习 2.D 则在相同模式基础上加入了小指。注意为了加入额外手指，节拍改为 5/4。记住：速度固然重要，但准确性更关键，并且要使用节拍器！当你熟悉这些模式后，尝试将它们应用到你喜欢的和弦形状上。

> ENG As you work through these exercises, you will quickly notice that your ring and little fingers are not as skilful as your thumb, index, and middle fingers. This is mainly because they share a ligament, which limits their dexterity. You can use the following exercise to enhance their strength and agility. While these two fingers may always feel a bit awkward, remember that precision is more important than speed. With consistent practice, you will see improvements over time. Although they may never match the strength and dexterity of your other fingers, we should still aim to make them as strong and agile as possible.

> 译 在练习过程中，你会很快发现无名指和小指的灵巧度不如拇指、食指和中指。这主要是因为它们共用一条韧带，限制了灵活性。你可以使用下面的练习来增强它们的力量与敏捷性。尽管这两个手指可能总会感觉有些笨拙，但请记住精确性比速度更重要。通过持续练习，你会逐渐看到进步。虽然它们可能永远无法与其他手指的力量和灵敏度相匹配，但我们仍应努力让它们尽可能强壮灵活。

> ENG Let's move on to start applying some notes on the fretboard. A common technique utilized with fingerpicking are hammer-ons and pull-offs. Exercise 2.F will get you started with this concept. The exercise is based around three basic open chord shapes: C major, G major, and D major. Once you can play the exercise comfortably, try playing it by using pull-offs instead. And I encourage you to try a combination of both.

> 译 接下来，让我们开始在指板上应用一些音符。指弹中常见的技巧是击音和勾音。练习 2.F 将帮助你入门。这个练习基于三个基本的开放和弦形状：C 大调、G 大调和 D 大调。当你能够舒适地演奏后，尝试改用勾音演奏。我也鼓励你尝试两者结合。

Finger Control & Plucking Chords > ENG Next, let's start plucking strings at the same time. This is an excellent approach for providing a different feel to your chords because strings are all plucked simultaneously vs a pick dragging (strummed) over the strings. It can give you better control over dynamics, better accuracy, as well as a different timbre.

> 译 接下来，我们开始同时拨奏多根琴弦。这是一种改变和弦感觉的好方法，因为所有琴弦是同时拨出的，而不是拨片拖弦。它可以让你更好地控制力度、更准确，并带来不同的音色。

> ENG To get started, exercise 2.G combines different finger parings (look at the note heads for which finger to use) paired with your thumb.

> 译 刚开始时，练习 2.G 结合了不同的手指搭配（看音符上的指法标记），与拇指配合使用。

> ENG Exercise 2.H expands on the previous exercise with the inclusion of another finger. Once you are comfortable with this, try adding more fingers until you are using all of them.

> 译 练习 2.H 在前一个练习基础上加入了另一个手指。一旦你熟练了，尝试继续加入更多手指，直到全部使用。

Test Yourself: Fingerpicking Exercises in Context > ENG Let's end this section with some exercises to put fingerpicking in context and give ourselves a little test.

> 译 最后，我们用一些练习来将指弹置于语境中，并给自己一个小测试。

> **ENG** Exercise 2.I demonstrates arpeggiating chords through fingerpicking, incorporating hammer-ons. While it is a simple example, it is an excellent starting point because this hammer-on technique is frequently used by math rock guitarists. Do not forget to incorporate pull-offs and a mix of both techniques! Keep your fingers positioned in chord shapes, ready for hammer-ons or pull-offs, as it will assist in executing the arpeggios smoothly. For cases where this approach does not fit, consider the optimal finger placement order to make it easier to play.

> **译** 练习 2.I 演示了通过指弹琶音和弦并加入击音。虽然这是一个简单的示例，但它是一个极好的起点，因为这种击音技巧在数学摇滚吉他手中经常使用。别忘了加入勾音以及两者的结合！保持手指在和弦形状中准备好击音或勾音，这将帮助你更顺畅地演奏琶音。如果这种方法在某些情况下不适用，请考虑最优的手指放置顺序以便演奏。

> **ENG** Exercise 2.J is an example of what you can do when plucking chords and arpeggiating them in the same arrangement. You will likely come across this way of playing in a variety of fingerpicked songs by math rock bands.

> **译** 练习 2.J 是一个示例，展示了在同一编排中拨奏和弦并对其进行琶音。如果你听过数学摇滚乐队的许多指弹歌曲，你很可能会遇到这种演奏方式。

> **ENG** Lastly, exercise 2.K is a nice post-rocky sounding example. The string skipping jump from the initial note in the first two bars would be a bit tricky if you were to use a pick but is made somewhat less challenging by using fingerpicking. To make this exercise easier, I recommend working out your finger placement first.

> **译** 最后，练习 2.K 是一个具有后摇风格的示例。前两小节中从起始音开始的跳弦如果用拨片演奏会比较棘手，但使用指弹后会稍微容易一些。为了让这个练习更容易，我建议先确定好手指位置。

3 Hybrid Picking / 混合拨弦

> ENG Hybrid picking is where we make use of a pick and fingers at the same time. It is not as prominent in math rock as other techniques we have explored but still very useful. I for one love this technique and employ it regularly. I like to use it because I can get the best of both worlds; the strengths of fingerpicking and having the option of using a pick at the same time. I implore you to learn this technique if you have not been doing so already.

> 译 混合拨弦是当我们同时使用拨片和手指的技巧。它在数学摇滚中的地位不如前面探索的技巧那样显著，但仍然非常实用。就我而言，我很喜欢这项技巧，并经常使用。它让我能够兼得两者优势：指弹的灵活性和拨片的力度。我强烈建议你学习这项技术，即使你还没有开始使用它。

> ENG This technique, when developed, can help you play fast precise licks, especially ones involving string skipping with a lot less effort. In addition, you can combine the pick with fingers for plucking chords (just like fingerpicking). Overall, it is a great compromise between using only fingers or only a pick.

> 译 当这项技巧熟练后，它可以帮助你演奏快速而精准的乐句，尤其是那些涉及跳弦的乐句，而且所需的努力更少。此外，你还可以将拨片与手指结合起来拨奏和弦（就像指弹一样）。总体而言，它是在只用手指和只用拨片之间的一个很好的折中方案。

Examples of guitarists who use hybrid picking

- > ENG Mario & Erick of CHON -Book

  > 译 CHON 的 Mario 与 Erick —《Book》

- > ENG Tim & Scott of Polyphia -Look but Don't Touch Playthrough

  > 译 Polyphia 的 Tim 与 Scott —《Look but Don't Touch》演奏版

- > ENG Alan of Tangled Hair -Yeah…It does look like a spider

  > 译 Tangled Hair 的 Alan —《Yeah…It does look like a spider》

> ENG Pay close attention to the singing parts and how all the strings resonate simultaneously when Alan plucks them, in contrast to when he strums during the louder sections. Additionally, starting from the two-minute mark, observe how Alan's use of hybrid picking enables him to execute that section, achieving the distinct sound it produces due to the accessibility afforded by this technique.

> 译 特别注意演奏中的连奏部分，以及当 Alan 拨奏时所有琴弦同时共鸣的效果，这与他在更响亮部分进行扫弦时形成了对比。另外，从两分钟处开始，观察 Alan 如何利用混合拨弦来演奏该部分，使其产生这种特殊音色，这正是该技巧所带来的便利性。

Hybrid Picking Dexterity Exercises > ENG Below you will find a series of exercises I have designed to help you become accustomed with the technique.

> 译 下面你会找到一系列我设计的练习，帮助你熟悉这项技巧。

> ENG To clarify, you will be holding a pick between your thumb and index finger (as you normally would) and using your remaining free fingers (middle, ring, and little finger) to pick notes. Compared to fingerpicking, I have noticed I rarely use my little finger when hybrid picking. That is not to say you will not need to use it, just a reminder that it is fine not to.

> 译 说明一下，你将把拨片夹在拇指和食指之间（像平常那样），并使用其余空闲的手指（中指、无名指和小指）拨奏音符。与指弹相比，我注意到混合拨弦时我很少使用小指。这并不是说你绝对不用它，只是提醒你不使用它也没问题。

> ENG To begin with, exercise 3.A is designed to get you started with the string skipping potentials of hybrid picking. I have labelled which fingers to use (remember Mac for this one). The lowest string will be played with your pick.

> 译 首先，练习 3.A 的设计是让你入门混合拨弦的跳弦潜力。我已标注出应使用的手指（记住 Mac 的指法）。最低音弦由拨片弹奏。

> ENG Keep your picking hand steady by resting it on the bridge, and notice how you hand and wrist are not jumping about like you would if you were to pick all these notes using only a pick.

> 译 将拨弦手稳稳放在琴桥上，观察你的手和手腕不会像仅用拨片弹奏时那样乱动。

> ENG I designed Exercise 3.B to get your fingers familiar with picking up and down in an arch like motion. This kind of arching melody motion is common in math rock.

> 译 练习 3.B 的设计是让你的手指熟悉类似拱形的上下拨动。这种拱形旋律运动在数学摇滚中很常见。

> ENG Bonus tip for exercise 3.B: once you are comfortable with playing it precisely, try picking up and down as fast as you can (without hurting yourself of course!). You can develop this into a quick raking like action across the strings (like sweep picking but with fingers included). It is very satisfying once you can do it!

> 译 练习 3.B 的额外提示：一旦你能精准演奏，尝试尽可能快地上下拨动（当然不要伤到自己！）。你可以把它发展成一种快速扫弦般的动作（类似扫弦，但加入了手指）。一旦你能做到，会非常有成就感！

> ENG Exercise 3.C is a slight variation on the previous exercise. It is designed to get your fingers moving in different patterns.

> 译 练习 3.C 是前一个练习的一个小变化。它的目的是让你的手指以不同模式移动。

> ENG Exercise 3.D applies a picking pattern along with some arpeggios.

> 译 练习 3.D 将拨弦模式与琶音结合起来。

> ENG Let's move on to plucking strings at the same time. Exercise 3.E gets you started with plucking two strings at a time using different finger combinations along with a pick. This may feel a little strange at first but that is completely normal.

> 译 接下来，我们开始同时拨奏多根琴弦。练习 3.E 帮助你开始练习使用不同手指组合配合拨片同时拨奏两根弦。一开始可能会有点奇怪，但这是完全正常的。

> ENG Exercise 3.F expands on the previous exercise by including the use of two fingers with a pick and in the last bar the inclusion of all three spare fingers!

> 译 练习 3.F 在前一个练习的基础上扩展到使用两个手指配合拨片，在最后一小节中加入了全部三个空闲手指！

> ENG Exercise 3.G uses just two notes from chord shapes (highest and lowest in this case) to create some interesting ideas.

> 译 练习 3.G 只使用和弦形状中的两个音（在这个例子中为最高音和最低音），以创造一些有趣的想法。

Test Yourself: Hybrid Picking Exercises in Context > ENG Exercise 3.H is an adaptation of the main chord progression to the song English Man in New York by Sting. Use your pick to play the lowest note of each chord and use your fingers to pluck the rest of the chord shape.

> 译 练习 3.H 是 Sting 的《English Man in New York》主和弦进行的改编。用拨片弹每个和弦的最低音，用手指拨奏剩余的和弦形状。

> ENG Lastly, exercise 3.I uses a repeating ascending pattern in triplets. Again, like exercises before, use the pick for the lower notes. Notice in the last bar, that by using hybrid picking, the string skipping becomes somewhat easier than trying to play it with just a pick.

> 译 最后，练习 3.I 使用三连音的重复上行模式。同前几个练习一样，用拨片弹低音。注意在最后一个小节，通过混合拨弦，跳弦变得比仅用拨片演奏更容易。

Part II

Beginner Music Theory, Chords & Progressions,

and Alternate Tunings

基础乐理、和弦进行与特殊调弦

4 Understanding Basic Music Theory / 基础乐理入门

> ENG In this section, I will provide an overview of music theory relevant to math rock and related music styles. We will explore fundamental principles of music theory, offering you a solid starting point to expand your knowledge. Before we dive in, you might have two essential questions:

> 译 在本节中，我将概述与数学摇滚及相关音乐风格相关的乐理知识。我们将探索音乐理论的基本原理，为你提供拓展知识的坚实起点。在深入之前，你可能会有两个关键问题：

- > ENG How Much Music Theory Knowledge is Needed to Learn, Write, and Play Math Rock?

  > 译 学习、创作和演奏数学摇滚需要多少乐理知识？

- > ENG What Music Notation Concepts Should I Know to Best Learn, Write, and Play Math Rock?

  > 译 为最好地学习、创作和演奏数学摇滚，我应该掌握哪些乐谱概念？

> ENG How Much Music Theory Knowledge is Needed to Learn, Write, and Play Math Rock?

> 译 学习、创作和演奏数学摇滚需要多少乐理知识？

- > ENG Key signatures (what sharps and flats are in each key: accidentals)

  > 译 调号（每个调中有哪些升降号：临时记号）

- > ENG Semitones and tones

  > 译 半音与全音

- > ENG Chords that are built from each note in a key (diatonic)

  > 译 从调内每个音建立的和弦（调式内和弦）

- > ENG Intervals (space between two notes)

  > 译 音程（两个音之间的距离）

- > ENG Chord progressions

  > 译 和弦进行

- > ENG Melody & harmony

  > 译 旋律与和声

- > ENG Rhythm (patterns of strong and weak pulses)

  > 译 节奏（强弱脉冲的模式）

- > ENG Time signatures

  > 译 拍号

- > ENG Major & minor scales

  > 译 大调与小调音阶

- > ENG Chord types - Triads, extended chords, sus chords, etc

  > 译 和弦类型——三和弦、扩展和弦、挂留和弦等

- > ENG Repetition and variation

  > 译 重复与变化

- > ENG Strumming patterns

  > 译 扫弦模式

> ENG What Music Notation Concepts Should I Know to Best Learn, Write, and Play Math Rock? (These are all related to reading music on a stave):

> 译 为最好地学习、创作和演奏数学摇滚，我应该掌握哪些乐谱概念？（这些都与五线谱阅读有关）：

- > ENG Time signatures (simple, compound, odd)

  > 译 拍号（简单拍、复拍、奇数拍）

- > ENG Key signatures

  > 译 调号

- > ENG Note divisions (whole, half, quarter, eighth, sixteenth, triplets, etc)

  > 译 音符时值（全音、二分音、四分音、八分音、十六分音、三连音等）

- > ENG Reading rhythm

  > 译 阅读节奏

- > ENG Rests

  > 译 休止符

- > ENG Ties and dots

  > 译 连线与附点

- > ENG Repetition signs

  > 译 重复记号

- > ENG Notation symbols for guitar: hammer-ons, pull-offs, slides, bends, staccato, tapping (left and right hand), - guitar pro is a great way to learn these notation symbols!

  > 译 吉他记号：击音、勾音、滑音、弯音、断音、点弦（左右手）——Guitar Pro 是学习这些记号的好工具！

- > ENG Dynamical markers

  > 译 力度记号

> ENG Learning these concepts is not as challenging as you might imagine, and the effort you invest in understanding them is highly rewarding. I recommend using this guide as a starting point and also obtaining a grade 1 music theory book, then gradually progressing through to grade 5. I myself have achieved a grade 5 in music theory, which is often suggested as the goal for those learning popular music styles. Beyond grade 5, music theory exams tend to focus more on classical music.

> 译 学习这些概念并不像你想象的那么困难，投入理解它们的努力也会带来丰厚的回报。我建议把本指南作为起点，同时准备一本 1 级音乐理论书，然后逐步学习到 5 级。我自己已获得音乐理论 5 级，这通常被认为是学习流行音乐风格的目标。超过 5 级后，音乐理论考试通常会更多地关注古典音乐。

Major and Minor > ENG Let's start with major and minor key signatures. Most modern math rock is quite melodic and usually uses chords and notes from one key signature (be this a conscious decision or not) and typically is comprised of harmony and melody you typically hear in popular music styles. This is somewhat different to earlier variations of math rock music, which is often referred to as being more dissonant in nature. I have linked an article below that was written by Nickk Hunter of Fecking Bahamas which clarifies these differences better than I can here.

> 译 让我们从大调和小调调号开始。大多数现代数学摇滚相当富有旋律性，通常使用来自一个调号的和弦与音符（无论这是有意为之还是无意识中形成），并且通常包含类似流行音乐的和声与旋律。这与早期的数学摇滚有所不同，早期作品常被认为更具不协和性质。我在下面链接了一篇由 Fecking Bahamas 的 Nickk Hunter 撰写的文章，比我在这里解释得更清楚。

> ENG Of course, this is not to say that every current math rock band and every song they write is more melodic than their predecessors. Even though it is a niche style of music there are many outliers to any claim that I could make.

> 译 当然，这并不是说每一支当前的数学摇滚乐队和他们写的每首歌都比前辈更具旋律性。尽管这是一个小众音乐风格，但对于我能提出的任何断言，都存在很多例外。

> ENG I will begin by explaining major and minor keys, what notes they consist of, and how you can create chords from these notes. Most importantly, I will demonstrate why understanding this basic concept can greatly impact your ability to generate song ideas.

> 译 我将从解释大调和小调开始，它们由哪些音符组成，以及如何从这些音符构建和弦。最重要的是，我会说明为什么理解这个基本概念会极大地影响你生成歌曲创意的能力。

Key Signatures > ENG A key signature consists of seven notes. Whether it is a major or minor key, a specific set of notes is used, which determines the sharps (#) or flats (b) for that key. The circle of fifths below shows the sharps and flats for each major and minor key.

> 译 调号由七个音符组成。无论是大调还是小调，它都会使用一组特定的音符，这决定了该调的升号（#）或降号（b）。下面的五度圈显示了每个大调和小调的升号与降号。

> ENG The outside of the circle (red) shows major keys, and the inside (green) shows minor keys. Notice how each major key has a relative minor key vice versa. For example, C major and A minor are relative of each other, meaning they share the same accidentals (which is no sharps or flats), same notes, and chords, but they start from different notes.

> 译 五度圈外圈（红色）表示大调，内圈（绿色）表示小调。注意每个大调都有一个关系小调，反之亦然。例如 C 大调和 A 小调是关系调，它们共享相同的升降号（此处为无升无降）、相同的音符和和弦，但起始音不同。

What Notes Are Available and How Many? > ENG When counting in sharps, we have these twelve notes: A, A#, B, C, C#, D, D#, E, F, F#, G, and G#. And when counting in flats, we have these twelve notes: A, Bb, B, C, Db, D, Eb, E, F, Gb, G, and Ab. It is important to note that A# and Bb (and others like C# and Db, D# and Eb) are the same notes. These are called enharmonic equivalents, and which one you use (sharps or flats) depends on your choice of key signature. A key signature will use seven of the twelve available notes. For example, C major uses C D E F G A B and back to C (the octave above).

> 译 按照升号计数，我们有这十二个音：A、A#、B、C、C#、D、D#、E、F、F#、G 和 G#。按照降号计数，我们有这十二个音：A、Bb、B、C、Db、D、Eb、E、F、Gb、G 和 Ab。需要注意的是，A# 和 Bb（以及 C# 和 Db、D# 和 Eb）实际上是相同的音。这称为等音，使用哪一个（升号还是降号）取决于你选择的调号。一个调号会使用这十二个音中的七个。例如 C 大调使用 C D E F G A B，然后回到高八度的 C。

> ENG Do not worry about learning all the keys right now, you will pick them up the more you use them, and you can always refer to the circle of fifths to know what sharps or flats are in the key you are using. To add clarity, I recommend watching this video about keys and the circle of fifths.

> 译 不要担心现在就学会所有调。你用得越多，就会越熟练，而且你随时可以参考五度圈，了解所用调中的升号或降号。为便于理解，我建议观看一段关于调和五度圈的视频。

> ENG Lastly, you may be asking 'how do I know which key I am in when writing or playing a piece of music?' I would like you to have a little think about how you could work this out, but I will show you a method soon!

> 译 最后，你可能会问“写歌或演奏时，如何判断自己处于哪个调?”我希望你先思考一下如何解决这个问题，但我很快会展示一种方法！

Why Is It Useful to Know Key Signatures? > ENG There are many benefits to knowing what key we are using when writing or playing songs. To start with, it is an excellent tool for knowing what chords, notes, and scales/arpeggios are available to us.

> 译 知道写歌或演奏时所用的调有许多好处。首先，它是了解我们可以使用哪些和弦、音符和音阶/琶音的极佳工具。

> ENG To elaborate, in the key of C major, we have these notes:

> 译 具体来说，在 C 大调中，我们有以下音：

> ENG 1 2 3 4 5 6 7 8 (1)

> 译 1 2 3 4 5 6 7 8 (1)

> ENG C D E F G A B C

> 译 C D E F G A B C

> ENG First off, if you consult the circle of fifths chart above and find C major, you can see there are no sharps or flats (accidentals) in the key of C major. Now you know the appropriate notes to use for a song or an idea in the key of C major (you could play any of the above notes and not hit a bum one). However, careful selection of notes based on music theory can tell us what notes will 'fit' nicely along with chords we choose, and which notes will not (this is not to say there are only correct and incorrect notes to use for a melody, but rather we will get different results from choosing different notes).

> 译 首先，如果你查看上方的五度圈并找到 C 大调，你会发现 C 大调没有升号或降号（临时记号）。现在你知道了在 C 大调中写歌或构思时可以使用哪些音（你可以弹奏上述任何一个音，不会出现难听的音）。然而，基于乐理的精心音符选择可以告诉我们哪些音更适合与所选和弦搭配，哪些音则不太适合（这并不是说旋律只有正确或错误的音，而是说选择不同的音会产生不同的效果）。

How Basic Chords Work > ENG By knowing the notes in a key signature, we can start to build chords. (For the sake of simplicity, we will stick to standard tuning).

> 译 通过了解调号中的音符，我们就可以开始构建和弦。（为简化起见，我们仍使用标准调弦。）

> ENG Continuing with the key of C major, let's build some basic triads (basic chords –those open chord shapes you first learn when learning guitar). We have the chords:

> 译 继续以 C 大调为例，构建一些基本三和弦（基本和弦——你学吉他时最先学到的那些开放和弦）。我们得到以下和弦：

> ENG C D E F G A B

> 译 C D E F G A B

> ENG Chords built from a major key follow this order: Major, Minor, Minor, Major, Major, Minor, Diminished. These are often referred to by roman numerals (see above). Upper case for major, and lower case for minor or diminished; I ii iii IV V vi vii°.

> 译 由大调构建的和弦遵循这个顺序：大和弦、小和弦、小和弦、大和弦、大和弦、小和弦、减和弦。通常用罗马数字表示（见上文）。大写表示大和弦，小写表示小和弦或减和弦；I ii iii IV V vi vii°。

> ENG Chords built in a minor key follow this order: Minor, Diminished, Major, Minor, Minor, Major, and Major or i ii° III iv v VI VII

> 译 小调构建的和弦遵循这个顺序：小和弦、减和弦、大和弦、小和弦、小和弦、大和弦、大和弦，表示为 i ii° III iv v VI VII。

> ENG Here is the key of A minor, which is the relative minor key of the key of C major (same chords but in a different order):

> 译 下面是 A 小调，它是 C 大调的关系小调（和弦相同但顺序不同）：

> ENG A B C D E F G

> 译 A B C D E F G

> ENG A chord needs to use at least three notes to be considered a chord, and the distances between the notes in a chord (intervals = spaces) makes it they type of chord it is. (Two notes played at the same time are referred to as intervals, but we could outline a chord with just two notes). Basic chords are made up of three notes, and you are probably already familiar with open chords such as C major:

> 译 一个和弦至少需要包含三个音，和弦中音符之间的距离（音程＝间隔）决定了它的类型。（同时演奏两个音称为音程，但我们也可以用两个音来概述一个和弦。）基本和弦由三个音组成，你可能已经熟悉像 C 大调这样的开放和弦：

> ENG The spelling for basic major chords is 1, 3, 5 - (Root, Major 3rd and Perfect 5th are the intervals counting from the C, the root note).

> 译 基本大三和弦的构成是 1、3、5（从根音 C 开始计数为根音、大三度和纯五度）。

> ENG If we refer to the C major scale, we can see where the notes that make up a C major chord come from. I have highlighted them in red.

> 译 如果我们参考 C 大调音阶，就可以看到组成 C 大和弦的音符来自哪里。我已将它们用红色标出。

> ENG The same applies for basic minor and diminished chords but their spellings are slightly different:

> 译 对于基本小三和弦和减三和弦也是同样的道理，但它们的构成稍有不同：

- > ENG Major: 1-3-5

  > 译 大和弦：1-3-5

- > ENG Minor: 1-b3-5 (flat 3rd)

  > 译 小和弦：1-b3-5（降三度）

- > ENG Diminished: 1-b3-b5 (flat 3rd, flat 5th)

  > 译 减和弦：1-b3-b5（降三度、降五度）

> ENG You can see this difference in action (the moving third degree and/or fifth degree) below:

> 译 你可以在下面看到这个差异的实际效果（移动的三度和/或五度）。

> ENG In a G major chord, there is a major 3rd (B), and when we change to the G minor chord: a flattened 3rd degree, a Bb. Lastly, in the diminished chord we have a flattened 3rd and a flattened 5th Db, and Bb. These intervallic differences are what makes these chords sound the way they do.

> 译 在 G 大和弦中，有一个大三度（B）；当我们转换到 G 小和弦时，三度变为降三度 Bb。最后，在减和弦中，我们有降三度和降五度 Db、Bb。这些音程差异就是让这些和弦听起来各不相同的原因。

How Do I Know Which Key Signature Is Used? > ENG Here is a method that I learned during my time studying music at university:

> 译 这是我在大学学习音乐时学到的一种方法：

> ENG If a key signature is not given, you can often figure out the key by analysing the harmonic elements such as chords, melody, and how different melodic instruments interact.

> 译 如果没有给出调号，通常可以通过分析和声元素来判断调，例如和弦、旋律，以及不同旋律乐器之间的互动。

> ENG For example, if you had an intro section of a song using chords: C major, F major, and G major, we can work out the key from this information. I would deduce that this song is in the key of C major, this is because these three chords are all contained in the key of C major, and because the F major and G major chords are right next to each other this tells me they are most likely to be the fourth and fifth chord of a major key (or the VI VII if we were in A minor):

> 译 例如，如果一首歌的前奏部分使用了 C 大调、F 大调和 G 大调，我们就可以从这些信息判断出这首歌的调。我会推测这首歌是 C 大调，因为这三个和弦都属于 C 大调，而且 F 大调和 G 大调紧挨着，这意味着它们很可能分别是大调中的 IV 和 V（如果是在 A 小调中，则是 VI 和 VII）。

> ENG C D E F G A B

> 译 C D E F G A B

> ENG Major Minor Minor Major Major Minor Dim

> 译 大小小大大小减

> ENG I ii iii IV V vi vii°

> ENG You can apply the same method if you see two minor chords together or two minor chords that occur together but are spaced out in the chord progression. For example, a C major -D minor -F major -E minor progression (I -ii -IV -iii), the D minor and E minor are spaced out in the progression, but they are next to each other alphabetically in the key of C major.

> 译 如果你看到两个小和弦并列出现，或者两个小和弦在和弦进行中分布较开，也可以应用同样的方法。例如 C 大调-D 小调-F 大调-E 小调的进行（I -ii -IV -iii），其中 Dm 和 Em 在进行中虽然分开，但在 C 大调的音阶中它们是相邻的。

> ENG Of course, this method is not bulletproof, especially when you have ambiguous information or less information to work with. For example, let's say you have just two chords: C major and G major, this could be either the key of C major or G major:

> 译 当然，这个方法并不绝对，尤其是当你得到的信息模糊或不足时。例如，假设你只有两个和弦：C 大调和 G 大调，这可能是 C 大调或 G 大调。

> ENG C D E F G A B

> 译 C D E F G A B

> ENG Major Minor Minor Major Major Minor Dim

> 译 大小小大大小减

> ENG I ii iii IV V vi vii°

> 译 I ii iii IV V vi vii°

> ENG G A B C D E F#

> 译 G A B C D E F#

> ENG Major Minor Minor Major Major Minor Dim

> 译 大小小大大小减

> ENG I ii iii IV V vi vii°

> ENG In cases such as these we generally need more information or take extra steps to work it out. First, I would check if there was an F# in the melody or a riff, this would tell us it is likely to be G major because that key contains an F#. Another way you can tell is by extending the chords to sevenths, you would either get a C major 7 and G dominant 7 (extended chords elaborated on in the following section) in the key of C major, or a G major 7 and C major 7 in the key of G major. I would try out both sets of chords and hear which set of chords fits the song better. From this, you can better deduce whether it was the key of G or C major. Of course, it could be either key, and it could be your decision to choose one or even make use of both!

> 译 在这种情况下，通常需要更多信息或进一步分析。首先，我会检查旋律或即兴片段中是否出现了 F#，如果有，那很可能是 G 大调，因为该调包含 F#。另一种判断方法是将和弦扩展到七和弦：在 C 大调中，你会得到 Cmaj7 和 G7（下节将详细讨论扩展和弦）；在 G 大调中，你会得到 Gmaj7 和 Cmaj7。我会尝试这两组和弦，听听哪一组更适合这首歌。由此，你可以更好地判断它是 G 调还是 C 调。当然，它也可能两者皆是，甚至你可以选择混用。

> ENG It is also worth keeping in mind that even though a song may be situated within a particular key, there may be chords and notes borrowed from other key signatures. There are methods we can use to help identify borrowed chords. And you will see how this is usually done in the borrowed chord segment towards the end of this section.

> 译 还应记住，尽管一首歌可能位于某个特定调中，但它可能会借用其他调的和弦和音符。我们有一些方法可以帮助识别借用和弦。这一点你将在本节末尾的借用和弦部分看到。

Task > **ENG** Below, I have given few chord progressions, and I would like you to work out the key of each one. The answers are provided at the end of this section. And keep in mind that some answers may be ambiguous or use more than one key. Fun times! Good luck!

> **译** 下面我给出一些和弦进行，请你判断每个进行所处的调。答案会在本节末提供。请记住，有些答案可能会存在歧义，或者涉及多个调。祝你玩得开心！好运！

1. > **ENG** E major -A major -G# minor -F# minor -E major

   > **译** E 大调-A 大调-G# 小调-F# 小调-E 大调

2. > **ENG** D minor -G7 -C major

   > **译** D 小调-G7 -C 大调

3. > **ENG** D major -E minor -F# diminished -G major

   > **译** D 大调-E 小调-F# 减和弦-G 大调

4. > **ENG** C major -G major -A minor -E minor

   > **译** C 大调-G 大调-A 小调-E 小调

5. > **ENG** G major -B minor -C major -C minor

   > **译** G 大调-B 小调-C 大调-C 小调

5 Chords and Chord Progressions / 和弦与和弦进行

> ENG By now, you should have an understanding of how basic chords are built and how they function in major and minor keys. The next step is to take these basic chords and make them into extended chords. Extended chords are used a ton in mathy and emo styles of music. They add a bit of extra flair to basic chords, so they have the potential to play with emotions and feelings in different ways.

> 译 到现在，你应该已经理解了基本和弦是如何构建的，以及它们在大调和小调中的功能。下一步是把这些基本和弦扩展成延伸和弦。延伸和弦在数学摇滚和情绪摇滚风格中大量使用。它们为基本和弦增加了一点额外的色彩，因此有能力以不同方式影响情感与氛围。

Extended Chords > ENG Extended chords are made by adding extra notes to basic chord spellings. Common extensions begin with the seventh chords. Again, like basic major, minor and diminished chords, extended chords have their own spellings. Below are the chord spellings for extended seven chords, the first possible extension:

> 译 延伸和弦是通过在基本和弦构成上添加额外音符来形成的。常见的扩展从七和弦开始。同样，像基本的大三和弦、小三和弦和减和弦一样，延伸和弦也有自己的构成方式。下面是延伸七和弦的和弦记号，这是第一个可能的扩展：

- > ENG Major 7th: 1-3-5-7

  > 译 大七和弦：1-3-5-7

- > ENG Dominant 7th: 1-3-5-b7

  > 译 属七和弦：1-3-5-b7

- > ENG Minor 7th: 1-b3-5-b7

  > 译 小七和弦：1-b3-5-b7

- > ENG Minor 7 flat 5: 1-b3-b5-b7

  > 译 半减七和弦：1-b3-b5-b7

> ENG You may have noticed that we have added two new chord types: dominant seven, and a minor seven flat five chord. Dominant chords have a strong sound to them and feel they need to resolve to another chord. M7b5 chords add more suspense to the basic diminished chord. They are useful as tension builders, and for slotting in-between chords in chord progressions for some suspenseful colouring.

> 译 你可能已经注意到，我们增加了两种新的和弦类型：属七和弦，以及半减七和弦。属七和弦有一种强烈的声音，给人一种需要解决到另一个和弦的感觉。m7b5 和弦为基本的减和弦添加了更多悬念。它们可作为张力构建器，并用于和弦进行中插入在两个和弦之间，以获得悬念感。

> ENG Beyond extended sevens, there are extended nines, elevens, and even thirteenth chords! Realistically it is up to you to choose which extended chords you want to learn and use. I recommend learning all the shapes

provided in this guide and over time find which work for you. For example, I found I largely prefer to use extended major and minor nines, and minor seven and minor eleven chords.

> 译 在延伸七和弦之外，还有延伸九和弦、十一和弦，甚至十三和弦！实际上由你决定想学习和使用哪些延伸和弦。我建议学习本指南中提供的所有形状，并随着时间推移发现哪些最适合你。例如，我发现我主要喜欢使用延伸的大九、小九，以及小七和小十一和弦。

Why Are the 9ths, 11ths, and 13ths Called That? > ENG Before we continue, you may be saying to yourself "hang on, there are only seven notes in a scale, so where are the 9ths, 11ths, 13ths coming from?" Let me elaborate. Counting through the scale of C major we eventually arrive at the octave (the same note but higher in pitch). If we continue counting past the octave, notes 2, 4, or 6 are referred to as 9th, 11th, and 13th degrees when used in chords. This is because they are extended above the octave.

> 译 在继续之前，你可能会问"等一下，音阶里只有七个音，九度、十一度、十三度是从哪里来的？"让我来解释。沿着 C 大调音阶数下去，我们最终会到达八度（同一个音，但更高的音高）。如果继续数过八度，第二个、第四个或第六个音在和弦中使用时会被称为九度、十一度和十三度。这是因为它们是高于八度的延伸音。

> ENG It is important to note that extended chords like 9, 11, and 13 are named that way only if they include the seventh degree. For instance, if you have a chord with the notes 1, 3, 5, and 9, it is simply called an Add9 chord, like Cmajadd9 for C major. The seventh degree is crucial for the sound of these extended chords.

> 译 重要的是要注意，只有当和弦包含七度时，九度、十一度、十三度这样的延伸和弦才会这么命名。例如，如果你有一个包含 1、3、5 和 9 的和弦，它只叫做 Add9 和弦，就像 Cmajadd9。七度对于这些延伸和弦的音色至关重要。

> ENG The reason I mention this is that with larger extended chords, you have more notes, but limited fingers and fretboard space to play them all. So, some "less important" notes are often left out to make it playable. However, it is noteworthy that the seventh degree is usually retained in these chords.

> 译 我之所以提到这个，是因为在更大的延伸和弦中，你有更多的音，但手指和指板空间有限，无法全部弹出。因此，为了可演奏性，有些"次要"的音常常会被省略。然而值得注意的是，七度通常会保留在这些和弦中。

Common Extended Chord Notes > ENG The following table shows how extended chords are often simplified for playability, with certain notes omitted while retaining the seventh degree.

> 译 下面的表格显示了延伸和弦如何为了便于演奏而简化，省略某些音但保留七度。

- > ENG Major 7: 1-3-5-7

  > 译 大七：1-3-5-7

- > ENG Major 9: 1-3-5-7-9, often dropping the 5th to become 1-3-7-9

  > 译 大九：1-3-5-7-9，通常省略五度变成 1-3-7-9

- > ENG Minor 11: 1-b3-5-b7-9-11, often dropping the 5th and 9th to become 1-b3-b7-11

  > 译 小十一：1-b3-5-b7-9-11，通常省略五度和九度变成 1-b3-b7-11

- > ENG Dom 13: 1-3-5-b7-9-11-13, often dropping the 5th and 11th (sometimes the 9th) resulting in 1-3-b7-(9)-13

  > 译 属十三：1-3-5-b7-9-11-13，通常省略五度和十一度（有时也省略九度），结果为 1-3-b7-(9)-13

Seventh Chord Voicings & Practice Progressions > ENG Below are extended seven chord voicings starting from the sixth, fifth, and fourth strings. Learning in all three positions will help with changing chords in progressions more smoothly and provide you with more options. So take your time to learn them, and keep in mind there are other ways of voicing these chords. These are shapes I like, and I think you will too.

> 译 下面是从第六、第五月和第四弦开始的延伸七和弦指型。学习这三种位置会帮助你更顺畅地在和弦进行中转换，并为你提供更多选项。所以慢慢学习它们，并记住还有其他方式来配位这些和弦。这些是我喜欢的形状，我想你也会喜欢。

> ENG In common music practice, changes between chords in a progression are usually accomplished by moving to the closest available voicing of a particular chord. There are a couple of reasons for this: smoothness of overall feel and less effort for the guitar player. However, interesting things can be achieved with bigger chord jumps, and this happens quite often in math rock. So do not take this as a rule that must always be followed.

> 译 在常见的音乐实践中，和弦进行中的变换通常通过移动到某个和弦的最近配位来完成。这有两个原因：整体感觉更平滑，以及吉他手的努力更少。然而，通过更大的和弦跳跃也可以实现有趣的效果，这在数学摇滚中经常发生。所以不要把这当作必须始终遵循的规则。

> ENG Here is a chord progression in the key of C major (CM7, Am7, G7, FM7) represented as I, vi, V, IV. I have shown three ways to play this progression on the guitar neck, starting from different initial chords. Try playing them using your preferred strumming patterns. Additionally, experiment with other ways to play the progression, possibly with larger chord jumps or spacing between the chords.

> 译 这是一个 C 大调中的和弦进行（CM7、Am7、G7、FM7），表示为 I、vi、V、IV。我展示了三种不同的方式在吉他指板上演奏这一进行，起始和弦不同。尝试用你喜欢的扫弦模式演奏它们。还可以尝试以其他方式演奏这个进行，比如更大的和弦跳跃或在和弦之间留出更多间距。

Ninth Chord Voicings & Practice Progressions > ENG Extended nine chords are hands down my favourite chord type to use. They have something about them that makes them just a bit more special than extended seventh chords. That being said, I almost always use a mixture of chord types in my chord progressions, because some chord shapes or chord types work better in a particular place in a progression than others.

> 译 延伸九和弦无疑是我最喜欢使用的和弦类型。它们有一种使它们比延伸七和弦更特别的感觉。话虽如此，我几乎总是在和弦进行中混合使用多种和弦类型，因为某些和弦形状或类型在进行的特定位置上比其他更合适。

> ENG Ninth chords have the spelling 1, 3, 5, 7, 9 and like extended seventh chords, I have included the spellings for each variation for clarity:

> 译 九和弦的构成是 1、3、5、7、9。和延伸七和弦一样，为了清楚起见，我也列出了每种变化的构成：

- > ENG Major 9th: 1-3-5-7-9

  > 译 大九：1-3-5-7-9

- > ENG Dominant 9th: 1-3-5-b7-9

  > 译 属九：1-3-5-b7-9

- > ENG Minor 9th: 1-b3-5-b7-9

  > 译 小九：1-b3-5-b7-9

- > ENG Minor 9 flat 5: 1-b3-b5-b7-9

  > 译 小九降五：1-b3-b5-b7-9

> ENG As you can see, the ninth degree of the chord is not altered (sharp or flat) in any of the extended ninth chords. The same is true for extended elevens and thirteenth chords. This is because the guide tones of a chord are its third and seventh. Even without a root you can almost completely define a chord by just knowing its third and seventh.

> 译 如你所见，在所有延伸九和弦中，第九度并没有被改变（升号或降号）。对于延伸十一和弦和十三和弦也是如此。这是因为和弦的导音是三度和七度。即使没有根音，仅凭三度和七度你也几乎可以完全定义一个和弦。

> ENG Note: The minor nine flat five is a little different because of its flattened fifth. It still contains the same guide notes as a minor nine chord: a flattened third and seventh, but until we add the fifth (either a natural fifth or a flattened fifth) into the chord we will not know for sure if the chord is a minor or a half-diminished chord (m7b5).

> 译 注意：小九降五与众不同，因为它有降五度。它仍然包含与小九和弦相同的导音：降三度和七度，但是在加入五度（无论是纯五度还是降五度）之前，我们无法确定该和弦究竟是小和弦还是半减和弦（m7b5）。

> ENG Another useful application of these chords is to substitute basic major, minor, or diminished chords or to use in conjunction with them for a more colourful sound.

> 译 这些和弦的另一个有用应用是替换基本的大三和弦、小三和弦或减和弦，或者与它们结合使用，以获得更丰富的音色。

Other Chord Variations Useful for Math Rock > ENG There are many more chords and chord types to explore beyond what we have discussed. Even without delving into altered chords and chord inversions, there is still a vast world of chords to discover. In this section, I will highlight some chords that I find particularly useful in mathy styles of music, and I will provide guidance on when and where to incorporate them.

> 译 除了我们已经讨论的内容之外，还有许多更多的和弦和和弦类型可以探索。即使不深入改变和弦和和弦转位，仍然有一个广阔的和弦世界可以发现。在本节中，我将重点介绍一些我认为在数学摇滚风格中尤其有用的和弦，并提供何时何地使用它们的建议。

Suspended Chords > ENG A suspended chord is a chord where the third degree has been omitted and usually replaced with either a major second or a perfect fourth. They have neither a major nor minor sound to them; rather, more of an "open sound" that often feels like it needs to be resolved.

> 译 挂留和弦是省略三度并通常用大二度或纯四度代替的和弦。它们既没有大调也没有小调的音色，而是一种更"开放"的声音，通常给人一种需要解决的感觉。

> ENG The most popular usage of suspended chords in math rock and Midwest emo is to embellish an existing chord in a progression, creating movement within the chord itself, or use them to situate the listener in a feeling of uncertainty. For example, if you were to play an A major chord followed by an Asus4 you will hear how this embellishes the original A major chord, giving it a different feel and flavour. In addition, you can play this backwards (Asus4 to A) for a more resolved sound.

> 译 挂留和弦在数学摇滚和中西部情绪摇滚中最受欢迎的用法是装饰和弦进行中的现有和弦，创造和弦内部的运动，或者将听众置于一种不确定的感觉中。例如，如果你先弹 A 大调和弦再弹 Asus4，你会听到它如何装饰原来的 A 大调，赋予它不同的感觉与色彩。此外，你也可以把它反过来弹（Asus4 到 A）以获得更解决感的声音。

> ENG You could even add an Asus2 to the progression. Listen how the slight change of the ascending higher note alters the overall feel of the chord.

> 译 你甚至可以在进行中加入 Asus2。听一听升高音的微小变化如何改变和弦的整体感觉。

> ENG Another useful application of these chords is to use them to replace dominant chords that sound out of place in a chord progression or to replace an unwanted dominant sound in a progression. Remember, dominant seven chords are built from the fifth degree (V chord) of a key. Try the progressions below to hear this in action.

Personally, I find the dominant chord sounds a little out of place for mathy sounds, but the suspended four chord replacement makes the progressions much more suitable for the desired sound I was looking for.

> 译 这些和弦的另一个有用应用是用它们替换在和弦进行中听起来不合适的属和弦，或者替换进行中不想要的属和弦声音。记住，属七和弦是从一个调的五度（V 和弦）构建的。尝试听下面的进行来感受这一点。就个人而言，我发现属和弦对数学摇滚的声音来说有点不合适，但用挂留四和弦替换后，使这些进行更适合我想要的音色。

> ENG One last point on suspended chords. It is worth pointing out that not every triad chord in a key can be replaced with either a sus2 or sus4 chord. I have made a handy chart below so you can see which chords can be substituted with a sus chord.

> 译 关于挂留和弦的最后一点值得指出：并不是调中的每个三和弦都可以被 sus2 或 sus4 替代。我做了一个实用的图表，让你可以看到哪些和弦可以用 sus 和弦替代。

Major and Minor 6 and 6add9 Chords > ENG Major and minor 6, add9, or 6add9 chords offer creative alternatives to extended chords while avoiding the seventh degree. They add harmonic interest without the complexity of including the seventh degree. Major 6 and 6add9 chords are often used because they are a great substitute for major seventh chords when it comes to providing a more resolved stable sound.

> 译 大/小六和弦、add9 和 6add9 和弦提供了比延伸和弦更有创意的替代方案，同时避免了七度。它们在不引入七度复杂性的情况下增加了和声趣味。大六和 6add9 和弦经常使用，因为它们在提供更解决、稳定的声音方面，是大七和弦的极佳替代。

> ENG The reason is that a major seventh chord contains a minor second interval between the root and the major seventh degree. Minor second intervals do not sound pleasant. If you play a major seventh chord, the interval is still there, but it is more spaced out and less obvious when the other notes are added. In contrast, major 6 chords and major 6add9 chords avoid this close interval and still produce a colourfully rich sound.

> 译 原因在于大七和弦包含根音与大七度之间的一个小二度音程。小二度听起来并不悦耳。如果你弹一个大七和弦，这个音程依然存在，但它更分散，并且在加入其他音符时不那么明显。相比之下，大六和弦和大六加九和弦避免了这个紧密音程，同时仍然产生丰富的色彩声音。

> ENG It is worth keeping in mind that minor 6 and minor 6add9 chords can have a more dissonant sound compared to their minor 7 counterparts. Therefore, it is a good idea to experiment with them to achieve the specific musical results you desire. Another useful application of these chords is to substitute basic major, minor, or diminished chords or to use them in conjunction with other chord types for a more colourful sound.

> 译 值得记住的是，与小七和弦相比，小六和小六加九和弦可能具有更加不协和的声音。因此，尝试它们是实现你想要的特定音乐效果的好办法。这些和弦的另一个有用应用是替换基本的大三、小三或减和弦，或者与其他和弦类型结合使用，以获得更丰富的色彩。

Extended Shell Chord Voicings > ENG Extended shell chord voicings simplify extended chords by including only the essential notes: the root, third, and seventh. These voicings capture the core tonal quality of the chord while removing some of the extra notes. They are particularly useful in ensemble settings with complex harmony and also work well with distortion, making them great for chord-based riffs.

> 译 延伸壳和弦配位通过仅包含必要的音：根、三度和七度，来简化延伸和弦。这些配位保留了和弦的核心音色，同时省略了一些额外音符。它们在复杂和声的合奏环境中尤其有用，也适合失真声音，是和弦型 riff 的理想选择。

> ENG Major seven shell voicings, minor seven shell voicings, and dominant seven shell voicings are all useful shapes to learn. Minor seven flat five shell voicings are the same as minor seven shell voicings, since the fifth is omitted in the shell voicing construction. This is not a problem; the shell voicing still functions as a seventh chord in the harmonic context.

> 译 大七壳配位、小七壳配位和属七壳配位都是值得学习的形状。半减七壳配位与小七壳配位相同，因为在壳配位构成中省略了五度。这不是问题；在和声语境中，壳配位仍然可以作为七和弦发挥作用。

> ENG Shell voicings are worth practicing alongside regular full-sized chords. Use them as a basis for crafting riffs or as a source of chord-writing ideas. They have a strong and emotive quality, especially with distortion or gain.

> 译 壳配位值得与常规全尺寸和弦一起练习。把它们用作创作 riff 的基础，或作为和弦写作的灵感来源。它们具有强烈而富有表现力的特质，尤其是在失真或增益下。

Chord Progressions & Mixing Chord Types > ENG A chord progression is a sequence of chords that forms the backbone of a song section. Some progressions sound more pleasing than others, and they can range from simple to highly intricate. Knowing the key helps you choose chords that will fit together well.

> 译 和弦进行是一系列和弦，构成歌曲片段的骨架。有些进行听起来比其他更悦耳，它们可以从简单到复杂不等。了解调有助于你选择能够和谐搭配的和弦。

> ENG For example, in the key of C major, the diatonic chords are: C, Dm, Em, F, G, Am, Bdim. These correspond to I, ii, iii, IV, V, vi, vii°. In a major key, the I chord has the strongest pull because it resolves the key center, while the IV chord is stable but tends to move to another chord. The vi chord is the relative minor and often has a strong emotional pull.

> 译 例如，在 C 大调中，调内和弦是：C、Dm、Em、F、G、Am、Bdim。它们对应 I、ii、iii、IV、V、vi、vii°。在大调中，I 和弦具有最强的解决力，因为它会回到调中心，而 IV 和弦则稳定但通常倾向于移动到另一个和弦。vi 和弦是关系小调，通常具有强烈的情感吸引力。

> ENG Below are several useful chord progressions based on common popular mathy and emo songs. These progressions are diatonic, meaning they only use chords from one key. They are not exact transcriptions of any particular song, but they capture useful harmonic functions.

> 译 下面是一些基于常见数学摇滚和情绪摇滚歌曲的实用和弦进行。这些进行都是调内的，也就是说仅使用一个调内的和弦。它们并非任何特定歌曲的精确转录，但它们抓住了有用的和声功能。

Functional Popular Diatonic Chord Progressions > ENG 1. IV –V –vi –I. This progression has a satisfying ascending feeling and resolves nicely to the tonic. It works well for an ending, a middle section, or a chorus.

> 译 1. IV –V –vi –I。这个进行有令人满意的上行感觉，最后很好地解决到主音。它适合作为结尾、中段或副歌。

> ENG 2. IV –ii –I –ii. This progression feels poppy with a touch of melancholy. It works well for a chorus-like section.

> 译 2. IV –ii –I –ii。这个进行听起来流行但带有一丝忧郁。它适合作为副歌式的段落。

> ENG 3. IV –V –vi –V. This progression is useful for building tension and feels like it needs to move somewhere. The vi chord often acts as an emotional anchor.

> 译 3. IV –V –vi –V。这个进行适合用来构建张力，听起来像是需要去往某处。vi 和弦通常起到情感锚点的作用。

> ENG 4. IV –ii –I. This progression works nicely for a verse or bridge, with a pleasant and stable feeling.

> 译 4. IV –ii –I。这个进行很适合作为主歌或桥段，带来愉悦且稳定的感觉。

> ENG 5. I –IV –vi. This progression has an uplifting ascending feel.

> 译 5. I –IV –vi。这个进行具有令人振奋的上行感觉。

> ENG 6. iii –IV –vi –V and 7. IV –V –iii –vi. These progressions have a math rock meets neo-soul vibe and work well with minor pentatonic or additional melodic colouring.

> 译 6. iii –IV –vi –V 和 7. IV –V –iii –vi。这些进行具有数学摇滚与新灵魂的混合风格，搭配小调五声音阶或额外的旋律色彩效果很好。

> ENG 8. ii –IV –vi –I. This progression has a suspenseful math-pop feel.

> 译 8. ii –IV –vi –I。这个进行具有悬念感的数学流行风格。

> ENG 9. I –iii –vi –ii. This is a nice progression for an epic outro.

> 译 9. I –iii –vi –ii。这个进行适合作为史诗般的结尾。

> ENG As a bonus, take a progression from Leon Russell's "Hummingbird" and adapt it for a minor key. The chord sequence i –VI –i –iv and i –V –VI –V has a bluesy original feel but can work well in a math rock context.

> 译 作为奖励，可以参考 Leon Russell 的《Hummingbird》并把它改编为小调。和弦序列 i –VI –i –iv 和 i –V –VI –V 原本有布鲁斯风格，但在数学摇滚语境中也非常适用。

Chord Progressions in Context (Standard Tuning) > ENG The following examples are derived from math rock and Midwest emo style songs. They are not exact reproduction, but they illustrate how many of the chords and progressions we discussed are used in real music.

> 译 以下示例来自数学摇滚和中西部情绪摇滚风格的歌曲。它们不是精确再现，但展示了我们讨论的许多和弦与进行如何在实际音乐中使用。

> ENG Midwest Emo, Progressive Math Rock, Japanese Math Rock, Emo Math Rock, Post-Hardcore/Math Rock all have their own characteristic progressions. Pay attention to the way the chords flow and the use of extended harmony.

> 译 中西部情绪摇滚、渐进数学摇滚、日本数学摇滚、情绪数学摇滚、后硬核/数学摇滚都有各自的特点。注意和弦的流动方式以及延伸和声的使用。

Chord Progressions Containing Borrowed Chords > ENG Borrowed chords are chords taken from another key than the parent key. They are often borrowed from the parallel minor or major. For example, if we are in C major, we could borrow chords from C minor. Borrowed chords add colour and unexpected movement to progressions.

> 译 借用和弦是从主调之外的另一个调借用的和弦。它们通常来源于平行小调或大调。例如，如果我们在 C 大调中，可以借用 C 小调的和弦。借用和弦为进行增添了色彩和意料之外的变化。

> ENG In the key of C major, the diatonic chords are I, ii, iii, IV, V, vi, vii°. In C minor, the chords are i, ii°, III, iv, v, VI, VII. Borrowing a chord from C minor when in C major introduces bIII, bVI, or bVII, which bring a darker or more dramatic character to the progression.

> 译 在 C 大调中，调内和弦是 I、ii、iii、IV、V、vi、vii°。在 C 小调中，和弦是 i、ii°、III、iv、v、VI、VII。当在 C 大调中借用 C 小调的和弦时，会引入 bIII、bVI 或 bVII，它们会为进行带来更暗沉或更戏剧化的特质。

> ENG Example borrowed progressions include:

- > ENG IV –I –ii –bIII: the borrowed major III adds an unexpected colour.

  > 译 IV –I –ii –bIII：借用的大三和弦增加了意外的色彩。

- > ENG I –ii –IV –bVII: the borrowed bVII replaces the diminished feel with a more stable major chord.

  > 译 I –ii –IV –bVII：借用的 bVII 用更稳定的大三和弦替代了减和弦的感觉。

- > ENG IV –I –iii –v: borrowing the minor v gives the progression a more characterful sound.

  > 译 IV –I –iii –v：借用的小五和弦为进行增添了更有特色的声音。

- > ENG vi –I –ii –v: another variation with borrowed v.

  > 译 vi –I –ii –v：另一种带借用 v 的变体。

- > ENG I –vi –bVI –ii: the borrowed bVI leads smoothly back into the progression.

  > 译 I –vi –bVI –ii：借用的 bVI 让进行顺畅地回到原路。

Tasks Answers > ENG Here are the answers for the chord key tasks:

> 译 以下是和弦调任务的答案：

1. > ENG E major –A major –G# minor –F# minor –E major

   > 译 E 大调–A 大调–G# 小调–F# 小调–E 大调

   =

   > ENG E major

   > 译 E 大调

2. > ENG D minor –G7 –C major

   > 译 D 小调–G7 –C 大调

   =

   > ENG C major

   > 译 C 大调

3. > ENG D major –E minor –F# diminished –G major

   > 译 D 大调–E 小调–F# 减和弦–G 大调

   =

   > ENG G major

   > 译 G 大调

4. > ENG C major –G major –A minor –E minor

   > 译 C 大调–G 大调–A 小调–E 小调

   =

   > ENG Either C major or G major (I–V–vi–iii in C or IV–V–ii–vi in G)

   > 译 可以是 C 大调，也可以是 G 大调（在 C 中为 I–V–vi–iii，在 G 中为 IV–V–ii–vi）

5. > ENG G major –B minor –C major –C minor

   > 译 G 大调–B 小调–C 大调–C 小调

   =

   > ENG G major with a borrowed iv chord (I–iii–IV–iv)

   > 译 带借用 iv 的 G 大调（I–iii–IV–iv）

6 Alternate Tunings / 特殊调弦

> ENG Alternative tunings are when you tune your guitar to something other than standard tuning (EADGCE). They are a key feature of math rock guitar and are often favoured by bands that write texturally rich guitar music.

> 译 特殊调弦是指将吉他调成与标准调弦（EADGCE）不同的音高。它们是数学摇滚吉他的一个关键特征，并且常被创作纹理丰富吉他音乐的乐队所喜爱。

Examples of Alternate Tunings > ENG This section will focus on two of the most common alternative tunings used: FACGCE and DAEAC#E (or CGDGBD, which is the same tuning two steps lower).

> 译 本节将重点讨论两种最常用的特殊调弦：FACGCE 和 DAEAC#E（或 CGDGBD，它其实是 DAEAC#E 降两度后的同一调弦）。

> ENG Some examples of FACGCE tuning in action include American Football –"Never Meant", Colossal –"Hot Probs", and TTNG –"Crocodile". Song examples in DAEAC#E include Sleepy Dog –"Screamers", Algernon Cadwallader –"Some Kind of Cadwallader", and Tiny Moving Parts –"Caution".

> 译 一些使用 FACGCE 调弦的实例包括 American Football 的《Never Meant》、Colossal 的《Hot Probs》以及 TTNG 的《Crocodile》。使用 DAEAC#E 调弦的歌曲示例包括 Sleepy Dog 的《Screamers》、Algernon Cadwallader 的《Some Kind of Cadwallader》和 Tiny Moving Parts 的《Caution》。

What Makes Certain Alternate Tunings Sound Great? > ENG You may be wondering "what makes some alternative tunings sound so damn nice?" The secret is in the note selection, and with a bit of simple music theory I can explain why.

> 译 你可能会想"是什么让某些特殊调弦听起来如此悦耳?"秘密在于音符选择，通过一些简单的乐理我可以解释原因。

> ENG If we look at the notes that make up standard tuning, E A D G B E, and the distance between the notes, we end up with a bunch of fourth intervals and one third interval. When strummed, this does not really sound like much. Technically, we could say the open strings of standard tuning form an Em11 chord.

> 译 如果我们查看构成标准调弦的音符 E A D G B E 及其间距，我们会得到一连串的纯四度音程和一个大三度音程。扫弦时，它听起来并没有太特别。从技术上讲，我们可以说标准调弦的空弦形成了一个 Em11 和弦。

> ENG If we take the tuning FACGCE and analyse it similarly, we will see why it sounds more like a chord than a random assortment of intervals. FACGCE contains more third intervals in its tuning, which makes it sound more like an Fmaj9 chord. This means the tuning is an open voiced chord tuning. It is based on a chord—in this case, an Fmaj9 chord. That is why it sounds so pleasing.

> 译 如果我们用相同的方式分析 FACGCE 调弦，我们就会明白为什么它听起来更像一个和弦而不是随机间隔的音。FACGCE 的调弦中包含更多的大三度音程，使它听起来更像一个 Fmaj9 和弦。这意味着这种调弦是一个开放和弦调弦。它基于一个和弦——在此例中是 Fmaj9 和弦。这就是它听起来如此动听的原因。

> ENG Now let's repeat the process for DAEAC#E tuning. This tuning is a bit more challenging to work out, but it is still based around a chord: an A/D chord (an A major chord with a D root, also known as a slash chord). This tuning is slightly less complex harmonically than FACGCE, but that lower D note over an A chord sounds

inspiring. The difference between these two tunings is that FACGCE contains a major seventh degree, which gives the tuning its characteristic happy-but-a-little-sad feel.

> 译 现在我们对 DAEAC#E 调弦重复这个过程。这个调弦稍微难以推导，但它仍然基于一个和弦：A/D 和弦（以 D 为根音的 A 大调和弦，也称作斜杠和弦）。与 FACGCE 相比，这个调弦在和声上略微简单一些，但低 D 音搭配 A 和弦听起来同样富有灵感。这两种调弦之间的区别在于 FACGCE 包含大七度音，这赋予了这种调弦一种快乐中带着一丝忧伤的特质。

String Gauges for Alternate Tunings > ENG If you are only temporarily changing strings or adjusting certain strings by a semitone, it is usually fine to stick with the same string gauge you use for standard tuning. However, for most alternative tunings, you will need more significant changes. I suggest using your regular string gauge to experiment with the new tuning, and if you decide to use it more frequently, explore changing individual string gauges accordingly.

> 译 如果你只是临时换弦或将某些弦调高/调低一个半音，通常仍然可以使用标准调弦时的弦规。然而，对于大多数特殊调弦，你将需要更明显的调整。我建议先用常用的弦规尝试新调弦，如果决定更频繁地使用它，再根据需要调整单根弦的弦规。

> ENG If you plan to use a tuning for an extended period or make larger tuning adjustments, I recommend:

> 译 如果你打算长时间使用某个调弦或进行较大的调弦变化，我建议：

1. > ENG For strings tuned down, use a thicker gauge string, usually one gauge higher than your current string.

   > 译 对于调低的弦，使用更粗的弦规，通常比当前的弦规大一号。

2. > ENG If you are tuning up, use a thinner gauge string.

   > 译 如果你要调高弦，则使用更细的弦。

3. > ENG For tuning changes larger than a whole tone, experiment with thicker or thinner gauges depending on direction.

   > 译 对于超过全音的调弦变化，根据调高或调低的方向尝试使用更粗或更细的弦。

> ENG This may require buying individual strings or hybrid sets, but it is worth it for better tension balance across the neck.

> 译 这可能需要购买单支弦或混合弦组，但为了更均衡的琴颈张力，这是值得的。

How to Write When Using Different Tunings? > ENG There is no easy answer or easy way of approaching writing in different tunings. I have a solid knowledge of where chords and notes are in standard tuning, but when I change to a different tuning, I lose that knowledge. You could look at this as off-putting, but I recommend seeing it as a good challenge.

> 译 在不同调弦中创作没有简单的答案或捷径。我对标准调弦中和弦与音符的位置有扎实的认识，但当我换到不同的调弦时，我就会失去这种认识。你可能觉得这令人却步，但我建议把它当作一个好挑战。

> ENG Exploring alternate tunings can also serve as excellent ear training. You will be challenged to find the notes you hear in your mind on the guitar, and this process can lead to discovering unique and appealing chords.

> 译 探索特殊调弦也可以作为极好的听力训练。你会被挑战在吉他上找到你脑海中听到的音符，这个过程可能会引导你发现独特而动听的和弦。

> ENG If you want to know where all the notes are in many different tunings, the tool https://fretboard-navigator.com/# is very helpful. You can choose a tuning, a key, and even scales to see where they are in a tuning on the fretboard. Bookmark it if you have not already.

> 译 如果你想知道不同调弦中所有音符的位置，工具 https://fretboard-navigator.com/# 会非常有帮助。你可以选择调弦、调和音阶，查看它们在指板上的位置。如果你还没有收藏，建议收藏一下。

> ENG Remember that open tunings offer a specific sound and feeling because they are tuned to a chord. You can use this to your advantage by incorporating the open tuning chord into your musical ideas and by including open strings within your chord choices. If you want to alter that feeling, use a capo or choose a different tuning instead.

> 译 记住，开弦调弦之所以提供特定的声音和感觉，是因为它们调成了一个和弦。你可以利用这一点，把开弦调弦的和弦融入你的音乐创意，并在和弦选择中包含开弦。如果你想改变这种感觉，可以使用变调夹或者选择另一种调弦。

Chords for FACGCE Tuning > ENG The approach to the following chords is twofold. First, I want to provide chords that capitalize on the sound of the tuning itself, making use of the open strings to create interesting open-string chord sounds within a key signature. Therefore, you will notice there is a chord for each note of the major scale. These chords sound wonderful clean or with light gain. To change key, apply a capo in the desired place. If using higher gain, you can also play the fretted notes and leave out the open strings for a tighter sound.

> 译 以下和弦的做法有两个方面。首先，我想提供利用调弦本身声音的和弦，利用开弦来在调号内创建有趣的开弦和弦音色。因此，你会发现每个大调音阶音都有一个和弦。这些和弦在干音或轻失真时都非常好听。要改变调，只需将变调夹夹在所需位置。如果使用更强的失真，你也可以演奏按住的音符而省略开弦，以获得更紧致的声音。

> ENG I also want to provide shapes that work well with higher gain so you do not need to learn two sets of chords. For example, for the Fmaj9 chord, play the top three strings if the full open shape is too muddy. With higher gain, these chords can become overly mushy if all open strings ring together, so be mindful of which strings are active.

> 译 我还想提供适合更高失真的和弦形状，这样你就不需要学习两套和弦。例如，对于 Fmaj9 和弦，如果完整的开弦形状太混浊，可以只弹最顶部的三根弦。使用更高的失真时，如果所有开弦一起响，和弦可能会变得过于糊状，所以要注意哪些弦处于激活状态。

FACGCE Chord Progressions in Context > ENG Below you will find chords and chord progressions from popular songs using FACGCE tuning. It is important to note that not all these chords are meant for strumming; some were originally played with picking patterns. However, they can still inspire your own ideas.

> 译 下面你会看到使用 FACGCE 调弦的流行歌曲的和弦与和弦进行。需要注意的是，并非所有这些和弦都适合扫弦；有些最初是用拨奏模式演奏的。但它们仍然可以激发你的创作灵感。

> ENG Examples include Covet – “Parachute”, Tiny Moving Parts – “You Lost Me”, Standards – “Smile”, American Football – “Never Meant”. Each progression demonstrates how the tuning can support open-string harmony and rich chord voicings.

> 译 示例包括 Covet 的《Parachute》、Tiny Moving Parts 的《You Lost Me》、Standards 的《Smile》、American Football 的《Never Meant》。每个进行都展示了这种调弦如何支持开弦和声与丰富的和弦配位。

FACGCE Arpeggios > ENG Below are major triad and minor triad arpeggio shapes for FACGCE tuning. The major shapes are based on F major, while the minor shapes are based on C minor. These shapes are movable, so you can adjust them based on the root of the major or minor chord you are using. Solid black notes on the charts show the root notes of each chord. Each chord has three arpeggios spanning an octave to cover all inversions: root position, first inversion, and second inversion.

> 译 下面是 FACGCE 调弦的大小三和弦琶音形状。大调形状基于 F 大调，而小调形状基于 C 小调。这些形状是可移动的，因此你可以根据所用大调或小调和弦的根音进行调整。图表中的实心黑点表示每个和弦的根音。每个和弦都有三个跨越一个八度的琶音，以涵盖所有转位：根音形、第一转位和第二转位。

> ENG As you study the charts, notice that the arpeggio shapes repeat at the octave. I have also marked where each arpeggio starts on each string. Although the root is positioned differently on each string, there are always three arpeggios: the root triad and the two possible inversions.

> 译 学习图表时，注意琶音形状在八度时会重复。我还标出了每个琶音在各根弦上的起始位置。虽然根音在每根弦上的位置不同，但始终有三个琶音：根音三和弦和两种可能的转位。

FACGCE Ideas in Context > ENG The following are five ideas I have written in FACGCE. I have included lots of noodling to get your brain working and to start noticing patterns and shapes available to you for writing licks in between your chords. Take your time with each one, and I recommend taking it phrase-by-phrase and building the idea until you can play it slowly. Examples two through five use a capo at the fourth fret.

> 译 下面是我用 FACGCE 写的五个创意。我加入了大量即兴片段，让你的思维开始运转，并开始注意可用于在和弦之间写乐句的模式与形状。每个例子都慢慢练习，我建议逐句学习，直到你能慢速演奏。示例二到五使用第四品的变调夹。

Chords for DAEAC#E Tuning > ENG Just like FACGCE tuning, I want to share chords that take full advantage of the unique sound DAEAC#E tuning offers. However, this tuning differs significantly from standard tuning, meaning it is much more difficult to find chords that align with every note in a key signature. So I take a slightly different approach by providing approachable chords and chord progressions that make the most of the open strings.

> 译 与 FACGCE 调弦一样，我也想分享充分发挥 DAEAC#E 调弦独特声音的和弦。然而，这种调弦与标准调弦差异很大，意味着更难找到与调号中每个音符完全对齐的和弦。因此，我采用稍微不同的方法，提供易于上手的和弦和和弦进行，充分利用开弦。

> ENG I also offer several chord progressions to illustrate how they can work together harmoniously. Try playing through these shapes and progressions to discover what you like. Then create your own progressions and explore riffs using the chord notes. These chords sound great cleanly or with a touch of light gain. If you want to use more distortion, you can play the fretted notes and exclude the open strings, but note that this may not work for every shape.

> 译 我还提供了几个和弦进行，说明它们如何和谐地协同工作。尝试演奏这些形状和进行，以发现你喜欢的声音。然后创建你自己的进行，并探索使用和弦音的 riff。这些和弦在干音或轻失真下都很不错。如果你想使用更多失真，可以只弹按住的音符并省略开弦，但注意这并不适用于每个形状。

DAEAC#E Chords and Progressions > ENG The tuning is used for several ideas, including “For the Sake of Brevity Style Chord Progression Idea”, “Night of The Pencils Chord Progression Idea”, “Upbeat Midwest Emo Style Chord Progression”, and “Descending Style Chord Progression with A Twist”. Some of these use a capo on the fifth fret.

> 译 这种调弦用于多个创意，包括“取悦简洁风格和弦进行”、“铅笔之夜和弦进行创意”、“欢快中西部情绪摇滚风格和弦进行”，以及“带转折的下行风格和弦进行”。其中一些在第五品使用变调夹。

DAEAC#E Chord Progressions in Context > ENG The chords and progressions below are derived from popular songs using this tuning. Consider them a foundation and use the examples to spark more ideas. Examples include Tiny Moving Parts –“Always Focused”, “Caution”, and “Dakota”, as well as Algernon Cadwallader –“Spit Fountain”, Sleepy Dog –“Screamers”, and Mountains –“Stride”.

> 译 下面的和弦与进行来源于使用这种调弦的流行歌曲。把它们视为基础，并用这些示例激发更多创意。示例包括 Tiny Moving Parts 的《Always Focused》、《Caution》和《Dakota》，以及 Algernon Cadwallader 的《Spit Fountain》、Sleepy Dog 的《Screamers》和 Mountains 的《Stride》。

DAEAC#E Arpeggios > ENG Here are major triad and minor triad arpeggios for DAEAC#E tuning. The major shapes are in A major, and the minor shapes are in F# minor. These shapes can be shifted based on the root of the chord you are using. Solid black notes indicate the root notes on each chart. Each chord has three arpeggios spanning an octave to cover root position and the two inversions.

> 译 这里是 DAEAC#E 调弦的大小三和弦琶音。大调形状是 A 大调，小调形状是 F# 小调。你可以根据所用和弦的根音移动这些形状。实心黑点表示每个图表中的根音。每个和弦有三个跨越一个八度的琶音，覆盖根音形和两种转位。

DAEAC#E Ideas in Context > ENG The final ideas are written in DAEAC#E and demonstrate how you can use the tuning to generate interesting melodic and chordal material. The examples are designed to get you exploring the unique voice of this tuning and to help you form ideas around the available open-string shapes.

> 译 最后的创意是用 DAEAC#E 写成的，展示了如何利用这种调弦生成有趣的旋律与和弦材料。这些示例旨在让你探索这种调弦的独特音色，并帮助你围绕可用的开弦形状形成创意。

Part III

Bonus Content

附加内容

7 Mastering the Fretboard / 指板精通

> ENG There are many methods out there to help you become familiar with note names and their locations on the neck: arpeggios, scales, movable chord shapes, chord shapes starting from different strings, the list goes on. I encourage you to try out different methods to find what works best for you. The following are methods I found rather useful for finding my way around the guitar neck more easily, more logically, and helped me on my way to mastering the fretboard.

> 译 有很多方法可以帮助你熟悉音名及其在指板上的位置：琶音、音阶、可移动和弦形状、从不同弦开始的和弦形状，等等。我鼓励你尝试不同的方法，找到最适合你的。下面是我认为非常有用的一些方法，能够帮助你更轻松、更有逻辑地掌握指板，走向指板精通。

Finding Notes Quickly —The Octave Method > ENG This method is a useful way to quickly locate notes all over the fretboard. It proves valuable for identifying root notes in chord shapes, arpeggios, scales, and more. I call it the "octave method," although it might have another name. Initially, focus on learning the positions of notes on the Low E and A strings (in standard tuning). While this may seem daunting if you are new to it, with practice, it becomes less challenging than you might expect. I have listed the notes starting from the open E and A strings up to their respective octaves on the twelfth fret (beyond the twelfth fret, you'll find the same notes but an octave higher).

> 译 这种方法是快速定位指板上音符的实用方式。它在识别和弦形状、琶音、音阶等的根音时非常有价值。我称它为“八度法”，虽然它可能还有其他名称。最初，重点学习标准调弦中低音 E 和 A 弦上的音符位置。对于新手来说这可能看起来有些困难，但通过练习，它会比你想象的容易。我列出了从空弦 E 和 A 到十二品对应八度的音符（超过十二品后，你会找到相同的音，但高一个八度）。

> ENG To illustrate how to apply this concept, I have provided a diagram. I have marked the note G on the low E and A strings with blue circles. Additionally, I have indicated the octaves of these notes (the same note but higher in pitch) with black circles. This can help you visualize how notes and their octaves are located on the guitar neck.

> 译 为了说明如何应用这个概念，我提供了一个示意图。我用蓝色圆圈标出了低音 E 和 A 弦上的 G 音，并用黑色圆圈标出了这些音的八度（相同音但更高）。这有助于你可视化音符及其八度在指板上的位置。

> ENG Notice how they have a particular spacing between each octave. From the G note (third fret) on the low E string, we jump a string (the A string) and find the octave G three frets higher (fifth fret on the D string). The same applies for the note G played on the A string. You may be familiar with these octave shapes if you have ever learned a pop punk or punk song. However, the spacing does change slightly when we start from the G note on the D string, and the same for the G note starting from the G string, the spacing is now four frets instead of three. The orange notes in the example below highlight this. I would not worry about getting this to stick in your mind because when played correctly you will hear them as octaves. If you play three fret spaces instead of four for octaves starting on the D and G strings you will get this major seven interval that clashes in sound, so you will instantly realize you need to slide your higher note up one fret to get the octave sound.

> 译 注意每个八度之间有特定的间距。从低音 E 弦上的 G 音（第三品）开始，我们跳一根弦（A 弦），在 D 弦第五品找到高八度的 G。A 弦上的 G 也有同样的关系。如果你学过流行朋克或朋克歌曲，你可能已经熟悉这些八度形状。然而，当我们从 D 弦上的 G 开始时，间距会略有变化；从 G 弦开始时，间距变为四品而不是三品。下

面示例中的橙色音符强调了这一点。不必过分担心记忆这些形状，因为正确演奏时你会听到它们是八度。如果你在 D 和 G 弦上以三品而不是四品来演奏八度，就会出现不协调的增七音程，你会立刻意识到需要把高音上移一个品位才能得到八度音。

> ENG Another handy trick is to notice that the low E and high E strings follow the same note order. Meaning you can mirror any note you play on either string to find the same note. I have indicated this with the red notes on the chart below.

> 译 另一个实用技巧是注意低音 E 弦和高音 E 弦具有相同的音序。这意味着你可以将任意一个 E 弦上的音在另一根 E 弦上镜像找到相同的音。我在下面的图表中用红色音符标出了这一点。

> ENG This system, I believe, is suited to players who have yet to have a built-up knowledge of note locations on the fretboard, so by following this method, you will slowly but surely start building your knowledge of note names and where they are located across the fretboard.

> 译 我认为这个系统适合那些还没有积累指板音符位置知识的玩家。通过这种方法，你会逐步建立起音名及其在指板上位置的认知。

Visualizing the Neck in Shapes - The Arpeggio Method > ENG By learning this method, you will be able to conjure up something both quickly and easily anywhere on the neck. Even something to play over fast changing or complex chord changes, and if time permitted you can really write something you will be proud of. Not only will it help you achieve the former, but it will also provide another useful approach to writing riffs, licks, melody lines, and so on. How and why this method will help you will become clearer as you work through this section.

> 译 通过学习这种方法，你将能够在指板任何位置快速且轻松地演奏东西。即使是在快速变化或复杂和弦变化之上也能奏出合适的内容，如果时间允许，你甚至可以真正写出令自己满意的乐句。它不仅帮助你实现上述目标，还为写作 riff、lick、旋律线等提供了另一种有用的方法。随着你阅读本节，这种方法如何帮助你以及为什么如此有用会变得更加清晰。

> ENG In this section, four types of extended arpeggios will be presented: major seven, minor seven, dominant, and minor seven flat five. Compared to arpeggio shapes you may have already learned; I will be taking a slightly different approach by teaching you arpeggios shapes that combine to form a pattern across the fretboard that you can easily learn. Also, I have included examples of how you could apply them.

> 译 本节将介绍四种扩展琶音：大七、小七、属七和半减七。与您可能已经学过的琶音形状相比，我将采用略有不同的方法，教你一些可以组合成指板上可学习模式的琶音形状。同时，我还会提供如何应用它们的示例。

> ENG Extended seven arpeggios give us an extra note to work with when compared with basic chord arpeggios (refer to the theory section for more detailed information) and will 'fit' over any iteration of a particular unaltered chord type. For example, a C major 7 arpeggio will work over a C major chord, C sus 2 or sus 4 chord, a C major 6 chord, C major 7 chord, C major 9 chord, and so on. If the chord or implied harmony is not altered, you should be safe to use it. A Cmaj7b5 chord is an example of an altered chord, the flattened 5th changes the quality of the chord, meaning it would NOT be harmonically sound if you were to play a Cmaj7 arpeggio along with it. Chords that contain notes such as b9, #9, b5 #5 are referred to as being altered chords, there is a lot to unpack there, but just think of it as making a chord sound more 'out there' in terms of sound.

> 译 与基本和弦琶音相比，扩展七琶音提供了额外的音符（详见乐理章节），并且可以适用于任何未改变性质的相同和弦类型。例如，C 大七琶音可用于 C 大和弦、Csus2、Csus4、C6、Cmaj7、Cmaj9 等。如果和弦或隐含和声没有被改变，你使用它通常是安全的。Cmaj7b5 是一个被改变的和弦示例，降五度改变了和弦的性质，这意味着如果你在它上面演奏 Cmaj7 琶音，就不会和谐。包含 b9、#9、b5、#5 等音的和弦被称为改变和弦，这里面内容很多，但你可以把它理解为让和弦听起来更“异样”的一种方式。

Major Seven Arpeggios Across the Fretboard > ENG Below you will find major seven arpeggios spanning one octave, meaning that when you get to the twelfth fret on your guitar, you should repeat the same patterns again. I have colour coded each shape to show how they are connected to each other and so you can see each individual arpeggio (chord) shape. Take your time working through each shape and find the most comfortable fingering, next, experiment with stringing them together into a chain, like in the transcribed version below, to help ingrain them into your memory and muscle memory.

> 译 下面你会看到跨一个八度的大七琶音，也就是当你到达第十二品时，应重复相同模式。我用颜色标识了每个形状，展示它们之间的连接，以便你看清每个单独的琶音（和弦）形状。慢慢练习每个形状，找到最舒适的指法，然后尝试将它们串联成一个链条，就像下面的转录版本那样，帮助你将它们记住并形成肌肉记忆。

> ENG Here is how they look transcribed for better clarity.

> 译 下面是它们的转录版本，便于更清晰理解。

> ENG Each individual chord shape used for the arpeggios are listed below. You can use any of these chord shapes interchangeably for major seven chords. Meaning, you will have to adjust the shapes when starting from a different root note, but the shapes will remain the same.

> 译 下面列出了每个琶音所使用的和弦形状。你可以将这些形状互换用于大七和弦。这意味着当你从不同根音开始时需要调整位置，但形状本身保持不变。

> ENG The following are a couple of examples of ideas you could potentially write. You could extend the examples here, but you will soon run into problems of things sounding too similar, thus these arpeggios are best suited to being combined with other arpeggios, scales and so on. At the end of this section, I will show examples of how you can combine a chain of arpeggios together for some musical phrases.

> 译 下面是几个你可能创作的示例。你可以延展这些示例，但很快你会发现听起来太过相似，因此这些琶音最适合与其他琶音、音阶等结合使用。本节末我会展示如何将一串琶音组合成音乐短句的示例。

Minor Seven Arpeggios Across the Fretboard > ENG Moving on to minor seven arpeggios, again they span one octave, meaning when you get to the twelfth fret, repeat the same patterns again. I have colour coded each shape to show how they are connected and so you can see each individual arpeggio (chord) shape. Take your time to work through each shape and find the most comfortable fingering for you, next, experiment with stringing them together into a chain like in the transcribed version below to help ingrain them into your memory and muscle memory.

> 译 接下来是小七琶音，同样跨一个八度，也就是到达第十二品时重复相同模式。我用颜色标识了每个形状，展示它们之间的连接，以便你看清每个单独的琶音（和弦）形状。慢慢练习每个形状，找到最舒适的指法，然后尝试将它们串联成一个链条，就像下面的转录版本那样，帮助你将它们记住并形成肌肉记忆。

Dominant Arpeggios Across the Fretboard > ENG Next, we have dominant arpeggios spanning one octave. Again, I have colour coded each shape to show how they are connected so you can see each individual arpeggio (chord) shape. Take your time to work through each shape and find the most comfortable fingering for you.

> 译 接下来是属七琶音，同样跨一个八度。我同样用颜色标识了每个形状，展示它们之间的连接，以便你看清每个单独的琶音（和弦）形状。慢慢练习每个形状，找到最舒适的指法。

> ENG Again, here are a few examples of short ideas you could potentially write. They sound a little odd in isolation, but they do work as individual licks or riff ideas. They sound better with the relevant dominant chord behind them (C7 for these examples), and the perhaps the following examples feel limited because they would work better in conjunction with a changing chord progression, meaning the arpeggios would change too. However, this is math rock so do whatever suits best!

> 译 再次给出几个你可能创作的简短示例。它们单独听起来可能有点奇怪，但作为独立的乐句或 riff 想法是可行的。它们与相应的属七和弦（例如本例中的 C7）搭配时会更好。或许下面的示例听起来有限，因为它们更适合与变化的和弦进行配合使用，也就是说琶音也会随之变化。不过这毕竟是数学摇滚，按你觉得合适的方式去做吧！

Minor Seven Flat Five (Half Diminished) Arpeggios Across the Fretboard > ENG Lastly, let us conclude with minor seven flat five arpeggios spanning one octave. Again, I have colour coded each shape to show how they are connected so you can see each individual arpeggio (chord) shape. However, I have changed the root note from C to D to better fit them in the examples. Take your time to work through each shape and find the most comfortable fingering for you. From there, experiment with stringing them together into chunks. The transcribed version below will aid in ingraining them into your memory and muscle memory.

> 译 最后，我们以跨一个八度的半减七琶音结束。我同样用颜色标识了每个形状，展示它们之间的连接，以便你看清每个单独的琶音（和弦）形状。不过我把根音从 C 改为 D，以便示例更契合。慢慢练习每个形状，找到最舒适的指法。然后尝试将它们串成片段。下面的转录版本将帮助你将它们记住并形成肌肉记忆。

Arpeggios - Examples Riffs & Licks > ENG Before moving forward, it is important that you are already familiar with the extended arpeggios we have just explored. In the upcoming section, the goal is to provide you with a few examples of how to combine different types of arpeggios to create riffs and licks. These can be played alongside a chord progression or used independently. These examples are intentionally designed to be relatively simple as this will make it easier for you to learn and grasp the concepts. As you work through them, try to identify which arpeggio shapes are being used.

> 译 在继续之前，重要的是你已经熟悉我们刚才探讨的扩展琶音。下一个部分的目标是为你提供一些示例，展示如何组合不同类型的琶音来创造 riff 和 lick。这些示例可以与和弦进行一起演奏，也可以独立使用。这些示例特意设计得相对简单，以便你更容易学习和掌握概念。练习时，尝试识别正在使用哪些琶音形状。

> ENG The following is an example of how you could write some ascending style licks. To make things easier notice the triplet groupings (grouping of three notes). Seeing it as groupings of three notes it will make it much easier on the fingers and your mind. Try them backwards too, for a descending feel.

> 译 下面是一个你可以写成上行风格乐句的示例。为了更容易理解，请注意三连音分组（每组三个音）。将其视为三音一组会让你的手指和思维更容易驾驭。也尝试反过来演奏，获得下行感觉。

> ENG Example 4G is a rather crude one, but I designed it to get you bouncing around the arpeggio shapes in two note groupings. Can you work out which arpeggio shapes the note groupings are derived from? Noticing them will help cement ideas in your memory better.

> 译 示例 4G 有些粗糙，但我设计它是为了让你在两音分组中在琶音形状间跳跃。你能找出这些音组来源于哪些琶音形状吗？注意它们会帮助你更好地将概念记住。

> ENG Example 4H takes the same idea but hashes it out into a passage. Sounds much more musical, right?

> 译 示例 4H 采用相同的想法，但将其扩展成一段乐句。听起来更富音乐性了，对吧？

> ENG Example 4I is another more musical idea. It is based around a IV, V, vi progression. You will hear the band Delta Sleep use this kind of chord progression quite regularly and to great effect in their music.

> 译 示例 4I 是另一个更具音乐性的想法。它基于 IV、V、vi 的进行。你会发现 Delta Sleep 乐队经常使用这种和弦进行，并且效果极佳。

> ENG Example 4J is another mathy riff sounding riff based round a I-IV progression. This time, I wanted to have quicker changes and some odd timing to make things more musical and an example of what you are likely to see in math rock styles of guitar. If you count the first bar as a grouping of three and four beats, and the second bar as three groupings of three it will make things easier.

> 译 示例 4J 是另一个具有数学摇滚风格的 riff，基于 I-IV 进行。这一次，我希望加入更快的变化和一些奇拍，以使乐句更具音乐性，并展示你在数学摇滚吉他风格中可能会看到的内容。如果你把第一小节看作 3+4 拍分组，第二小节看作三个 3 拍分组，会更容易理解。

> ENG Lastly, I could not help but include a couple of tapping examples. This is math rock after all!

> 译 最后，我忍不住也要包含几个点弦示例。毕竟这是数学摇滚嘛！

> **ENG** As time passes, and the more time you dedicate to these methods, you will better internalize where notes are on the fretboard and formulate your ideas to greater effect. This concluded this segment. Next, we will look at functional strumming patterns.

> **译** 随着时间推移，你对这些方法投入的时间越多，就会越能内化指板上音符的位置，并更有效地形成创意。至此，本节结束。接下来，我们将看看功能性扫弦模式。

8 Functional Strumming Patterns / 功能性扫弦模式

8.1 Overview / 概述

> **ENG** Strumming patterns are the foundation of rhythm guitar in math rock. Unlike traditional rock or pop, math rock often employs odd time signatures and syncopated rhythms.

> **译** 扫弦节奏是数学摇滚节奏吉他的基础。与传统摇滚或流行不同，数学摇滚常用奇数拍号和切分节奏。

EN Let's start with common time signatures. **CN** 从最常用的拍号开始。

核心术语

- 奇数拍号（*Odd Time Signature*）
- 切分音（*Syncopation*）
- 连音符（*Tuplet*）

提示

数学摇滚扫弦常强调反拍。练习时先用节拍器慢速感受重音移位。

8.2 3/4 Time / 四三拍

3/4 扫弦模式

3/4 拍中常见的扫弦模式：

```
e|-----------------|-----------------|
B|-----------------|-----------------|
G|-----------------|-----------------|
D|-------2---------|-------2---------|
A|-----3---3-------|-----3---3-------|
E|---3-------3-----|---3-------3-----|
```

> **ENG** This pattern uses palm muting on the downbeats with open strings ringing.

> **译** 这一模式使用下拍制音，空弦音延续。

练习 8.1 基础扫弦

以 BPM = 80 练习上述模式：

1. 保持手腕放松
2. 强调第 2 拍反拍
3. 先慢后快，逐步加速

8.3 4/4 Time / 四四拍

> ENG In 4/4, math rock strumming creates tension by emphasizing the & of 2 and & of 4.

> 译 在 4/4 拍中，数学摇滚扫弦通过强调第 2、4 拍反拍制造张力。

Functional Strumming Patterns in 4/4 > ENG You might have heard strumming pattern 5A or something similar at the beginning of or in the breakdown section of math rock/Midwest emo songs. It is a common pattern but usually has slight variations. In example 5B, the second strum is longer and the last two notes should be played as dead notes (muted strings).

> 译 你可能在数学摇滚/中西部情绪摇滚歌曲的开头或过门部分听过 5A 或类似的扫弦模式。这是一个常见的模式，但通常会有细微变化。在示例 5B 中，第二次扫弦更长，最后两个音应作为死音（哑音）的弹奏。

> ENG Strumming pattern 5C has a short staccato first strum followed by regular eighth-note strums. Adding dead notes before each phrase gives an interesting feel, as shown in 5D. Example 5E often appears with palm muting as a section builds toward something.

> 译 5C 模式以简短的断音第一扫开始，随后是常规四分之一音符的扫弦。在每个乐句前加死音会产生有趣的感觉，如 5D 所示。示例 5E 通常出现在用掌制音时，作为一段音乐向高潮推进。

> ENG Strumming pattern 5F is great for math-pop ideas. It has a poppy groove with syncopation that makes it less conventional than typical pop. You can also play it without the rests for a slightly different feel.

> 译 5F 模式非常适合数学流行风格。它具有流行感的律动，加上切分，使其相比普通流行更不寻常。你也可以不加休止符地演奏它，感觉会略有不同。

> ENG Strumming pattern 5H fills in all the gaps while retaining characteristic syncopation. It works well for an upbeat intro. Pattern 5I has a lot of energy and can be used to great effect in verses or bridges, especially if you palm-mute some intervals. Example 5J replaces some strummed notes with dead notes for an even more upbeat feel.

> 译 5H 模式填满了所有空隙，同时保留了典型的切分感。它很适合作为欢快的开头。5I 具有很强的能量，可在主歌或桥段中大放异彩，尤其是当你对某些音进行掌制音时。示例 5J 用死音替换了一些扫弦音，听起来更有活力。

Two-Guitar Strumming Patterns > ENG Let's move on to functional strumming patterns for two guitars. Example 5K is based around a pattern used by Meet Me In St. Louis. The alternating pattern is the same for the first two beats, then it varies from beat three onwards. Panning the parts left and right with this kind of idea sounds excellent.

> 译 接下来是适合两把吉他的功能性扫弦模式。示例 5K 基于 Meet Me In St. Louis 使用的一个模式。前两拍的交替模式相同，从第三拍开始则会变化。这样的想法在左右声道分开时听起来非常棒。

> ENG You could play the pattern on one guitar, but you will miss the separation effect of two guitars. Example 5L shows a groove where one of the beats is syncopated, typically the last beat of the bar. The first bar uses a unified guitar pattern and the second bar alternates between two guitars.

> 译 你可以在一把吉他上演奏这个模式，但你会错过两把吉他分离的效果。示例 5L 展示了一个律动，其中一个拍通常是最后一个拍的切分。第一小节使用统一的吉他模式，第二小节则在两把吉他之间交替。

> ENG Finally, pattern 5M is excellent for creating space in a song while maintaining an upbeat feel with staccato eighth notes in one guitar. You can experiment with the placement of the full-chord guitar, or even play the second guitar part as arpeggios for contrast. Since the chord is only strummed once per bar, it is a good place to add reverb, delay, or modulation to fill out the sound.

> 译 最后，5M 模式非常适合在歌曲中创造空间，同时一把吉他的断音八分音符仍然保持欢快感。你可以尝试调整全和弦吉他的放置位置，甚至可以让第二把吉他演奏琶音以形成对比。由于每小节只扫弦一次，这是一个适合添加混响、延迟或调制以丰富音色的好机会。

8.4 3/4 Time / 四三拍

> ENG Moving on to 3/4 time. This is one of my favourite time signatures (along with 7/8). They are a great way to inject energy into a song because the bar changes usually feel quicker due to the shorter length.

> 译 接下来是 3/4 拍。这是我最喜欢的拍号之一（还有 7/8）。它们是给歌曲注入能量的好方法，因为小节更短，听起来变化更快。

> ENG The following are some common 3/4 strumming pattern variations you will hear in math rock bands. Example 6A accents beat three. This idea is taken from the song "Pretend" by Via Luna. If you really accent the final strum, it is a lot of fun.

> 译 下面是一些在数学摇滚乐队中常见的 3/4 扫弦模式变体。示例 6A 强调第三拍。这个想法来自 Via Luna 的"Pretend"。如果你真正强调最后一扫，会非常有趣。

> ENG Strumming pattern 6B takes the same idea but arpeggiates the chord in the same pattern, keeping the accent on beat three. Pattern 6C is inspired by Totorro's "Home Alone" and is a fantastic driving pattern that works well in most songs.

> 译 6B 模式采用相同的想法，但在相同模式中拨奏和弦，仍然保持第三拍的重音。6C 模式则受到 Totorro 的"Home Alone"启发，是一个出色的推进型模式，适用于大多数歌曲。

> ENG Pattern 6D uses the same idea and shows how it can be adapted to 5/8 time for an interesting variation. You can count the bar as 1 & 2 & 3 or as 5 eighth notes to keep the feel of 3/4 with an abrupt ending. Pattern 6E is inspired by Blakfish's "Preparing for Guests" and injects energy into a song. Pattern 6F takes the same idea and completes the phrase with an arpeggio.

> 译 6D 模式采用相同的思路，并展示了如何将其改编到 5/8 拍以获得有趣的变化。你可以把小节计为 1 & 2 & 3，或者作为 5 个八分音符，以保留 3/4 的感觉同时有一种突兀的结束。6E 模式受到 Blakfish 的"Preparing for Guests"启发，为歌曲注入能量。6F 模式则用琶音完整地结束乐句。

8.5 5/8 Time / 五八拍

> ENG Now, let's delve into an unusual time signature: 5/8 time. I included this because it is a logical progression from 3/4 time. In 3/4, we count three quarter notes. In 5/8, there are five eighth notes. If we subdivide 3/4 into eighth notes, it becomes six eighth notes. Thus 5/8 can feel like a shortened bar of 3/4.

> 译 现在，让我们深入探讨不寻常的拍号：5/8 拍。我之所以包含它，是因为它是从 3/4 拍的一个合理延伸。在 3/4 拍中，我们数三个四分音符。在 5/8 拍中，则是五个八分音符。如果把 3/4 拆分成八分音符，就变成了六个八分音符。因此 5/8 会感觉像是一个缩短的 3/4 小节。

> ENG You can think of 5/8 as 1 & 2 & 3, with the absence of the & of 3 giving the bar a cut-short feel. The example videos will help if the rhythm is hard to work out. In 5/8, sixteenth notes are noted with + symbols for clarity.

> 译 你可以把 5/8 视为 1 & 2 & 3，缺少第三拍的 & 会让小节感觉被截短。如果节奏难以掌握，可以看示例视频。为了清楚起见，5/8 中的十六分音符用 + 符号表示。

> ENG It is common to accent the 1st and either the 3rd or 4th beats in 5/8. Pattern 7A accents 1 and 3. Pattern 7B applies palm muting between those accents. Pattern 7C accents 1 and 4, and 7D adds another accent on beat 5. Pattern 7E combines palm muting with 1 and 4 accents, while 7F introduces more space and is an excellent way to make 5/8 feel breathable.

> 译 在 5/8 中，通常会强调第一拍和第三拍或第四拍。7A 模式强调一和三。7B 模式在这些重音之间加入掌制音。7C 模式强调一和四，而 7D 在第五拍上再加一个重音。7E 结合了掌制音和一、四拍重音，而 7F 则引入更多空间，是让 5/8 听起来更通透的绝佳方式。

A　术语表 / Glossary

English	中文	说明
Alternate tuning	特殊调弦	非标准调弦方式
Arpeggio	琶音	将和弦音依次弹出
Downstroke	下拨	拨片向下拨弦
Gallop rhythm	马蹄节奏	前八后十六的节奏型
Hybrid picking	混合拨弦	拨片 + 手指同时使用
Odd time signature	奇数拍号	如 5/4、7/8、11/8 等
Open tuning	开放调弦	空弦即可构成和弦
Palm mute	制音	手掌轻触琴弦抑制振动
Polyrhythm	复合节奏	同时使用两种不同节奏
Pull-off	勾弦	按弦手指离开弦发出音
Syncopation	切分音	强调弱拍或非重音位置
Tapping	点弦	用手指敲击指板发声
Tuplet	连音符	一拍均分为不等份
Upstroke	上拨	拨片向上拨弦